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Modelling near electrode regions of an atmospheric-pressure arc discharge

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Abstract

A unified one-dimensional model of the near-electrode regions of an atmospheric-pressure argon arc is presented. A single set of equations, including species continuity, electron and heavy-particle energy equations and Poisson's equation, is solved from the electrode surface, through the space-charge sheath and the non-equilibrium layer, to the bulk plasma, whose state is supplied by measurements or by an LTE model of the arc. The cathode surface temperature follows from a heat balance with thermionic emission. For a high-intensity arc, the structure of the cathode and anode layers, the electron energy balance and the heat fluxes to the electrodes are presented. The collisionality of the sheaths is examined. The electrons cross both sheaths almost without collisions, and their density follows the Boltzmann relation, which the drift-diffusion description reproduces. The ions are collisional, more so at the anode, and do not reach the Bohm velocity at the sheath edge, so that a Bohm flux overestimates the ion flux. The dependence of the anode layer on the bulk temperature and the current is also presented. Finally, the responses of both layers to changes of the arc current and of the bulk temperature are computed, to determine how slowly these must change for the layers to be treated as quasi-steady.

1 Introduction

Arc discharges are widely used in welding, cutting and thermal spraying, and in recent years they have found new applications in plasma pyrolysis, advanced material synthesis and metallurgical reduction [1, 2, 3, 4]. A typical arc discharge consists of a hot quasi-neutral plasma, referred to as the bulk plasma, which is millimeters to centimeters in size and in which convection, thermal conduction, Joule heating and radiation interact under conditions often close to local thermodynamic equilibrium (LTE). Over a region of a millimeter from the electrodes, which is often referred to as the near-electrode or electrode region/layer, the interaction with the electrodes results in a highly non-uniform plasma with strong gradients in the species densities and temperatures. Very close to the electrodes (at sub-micrometer distances), charge separation develops, quasi-neutrality breaks down and strong electric fields arise.

Because of the high temperatures (∼20 000 K) and very small length scales compared with macroscopic dimensions, analyzing an arc discharge through experiments is difficult, and therefore there is a strong dependence on computational models. Computational models solve a set of governing equations and boundary conditions to describe a plasma phenomenon. Plasma phenomena are, however, characterized by a wide range of length scales in arc discharges. Resolving these wide range of length scales over the entire discharge is neither computationally practical nor necessary for most applications. For example, Poisson's equation, which governs the charge separation and the electric field near the electrodes, need not be solved in the bulk plasma, where quasi-neutrality holds. Conversely, the electrode regions are thin compared with the flow and geometric scales of the arc, so convection, electrode jets and the electrode geometry can be neglected there. These layers are governed by diffusion, reactions and sheath effects. Arc models therefore typically combine a quasi-neutral model of the bulk plasma with a separate near-electrode model. Several strategies have been adopted for modelling the bulk plasma, and the choice determines what the near-electrode model must describe.

(i) Local thermodynamic equilibrium. Measurements of the species temperatures and the electron density indicate that the arc core is generally close to LTE [5, 6, 7]. In an LTE model, electrons and heavy particles are assumed to have a single temperature T with Maxwellian velocity distributions and Boltzmann excitation. The plasma state and its properties are determined from the Saha equation and Dalton's law. The governing equations typically consist of mass, momentum and energy conservation together with current continuity [8, 9, 10, 11]. These models are computationally efficient because the species densities and transport properties need not be solved for and follow from the local temperature and pressure. An LTE bulk model, however, requires a near-electrode model that captures the separation of the electron and heavy-particle temperatures, the departure from Saha equilibrium, the breakdown of quasi-neutrality and the formation of the space-charge sheath [12, 13, 14].

(ii) Two-temperature models. Two-temperature (2T) models treat electrons and heavy particles as separate Maxwellian populations at temperatures T_e and T_h, coupled through elastic energy exchange. In two-temperature equilibrium (2T-EQ) models, the composition is, in addition, obtained from a generalized Saha relation for $T_e \neq T_h$. Hsu and Pfender [15] showed that $T_e \approx T_h$ in the high-density arc core, whereas the temperatures separate significantly in the low-density fringes and near the anode. The generalized Saha closure is not unique, however, and different formulations can predict electron densities that differ by up to an order of magnitude once T_e/T_h departs from unity [16]. Freton *et al.* [17] have further shown that several published formulations distribute the ionization term and the reactive thermal conductivity inconsistently between the electron and heavy-particle energy equations. A 2T bulk model still requires a near-electrode model that captures the ionization layer, or presheath, where Saha equilibrium is violated, as well as the breakdown of quasi-neutrality and the formation of the space-charge sheath [18, 19].

(iii) Nonequilibrium models. Fully nonequilibrium models abandon equilibrium closures such as LTE and Saha equilibrium and let the plasma composition and the electron and heavy-particle temperatures evolve independently. They typically include multicomponent species transport and separate electron and heavy-particle energy equations. Examples are the nonequilibrium arc models of Haidar [20], Gonzalez *et al.* [21] and Amakawa *et al.* [22]. Because they resolve departures from equilibrium directly, these models can describe the transition from the LTE-like bulk plasma to the strongly nonequilibrium electrode region. The space-charge sheath, however, is still treated separately, by a sheath model matched to the quasi-neutral plasma [23].

A widely used near-electrode modelling approach treats the ionization layer, or presheath, as a collisional region and couples it to a collisionless sheath model [24, 25, 19, 18]. This avoids solving the stiff Poisson equation across the sheath. Almeida, Benilov and Naidis [12], however, estimated the number of collisions that an ion undergoes while crossing the cathode space-charge sheath of an atmospheric-pressure argon arc and found it to be of order unity rather than small, so that, strictly speaking, the collisionless-sheath assumption is not justified. A kinetic (particle-in-cell/Monte Carlo collision) study of the near-cathode region by Sun *et al.* [26] likewise supports this view. At the anode, however, Semenov *et al.* [14] showed that the common collisionless sheath model, in which the ion velocity at the sheath edge is set to the Bohm velocity, fails. A physically consistent electrode model for high-pressure arcs should therefore account for collisional ion transport within the sheath.

The alternative is *unified modelling*, in which a single set of governing equations, including Poisson's equation, is solved from the electrode surface through the sheath to the bulk plasma, with no matching between sub-regions [12, 14, 26, 27]. Ion collisions in the sheath are then accounted for by construction. Because a unified model captures both thermal and ionization nonequilibrium, it can be matched directly to an LTE model of the bulk plasma. In addition, this approach requires no prior knowledge of the direction of the electric field. Field reversals, and changes in the sign of the potential drop near the anode, have been reported in atmospheric-pressure arcs, depending on factors such as the attachment mode, the current density and electrode ablation [28, 29, 30, 23]. When these equations are solved self-consistently across the near-electrode region, the potential structure of the sheath and its polarity emerge from the solution instead of being assumed a priori. Unified models are, however, computationally expensive, because the mesh must resolve the very small length scales of the sheath. Consequently, their application to arc discharges has largely been restricted to one-dimensional calculations extending from the electrode into the bulk plasma [12, 13, 31, 14].

In this paper we adopt the decomposition described above, treating the quasi-neutral bulk plasma and the near-electrode region with separate models (figure 1), and present a transient one-dimensional unified near-electrode model. The near-electrode model is unified in this sense, with drift-diffusion fluxes for the charged species coupled to Poisson's equation. It determines the

charged-particle distributions, the species fluxes, the sheath voltage drop and the heat flux to the electrode simultaneously. The same system of equations is solved continuously from the quasi-neutral bulk plasma at one end of the domain, through the near-electrode transition region, and into the sheath. In previous one-dimensional unified models [12, 31, 13, 14], the bulk-plasma state was obtained from the asymptotic arc column, assuming a uniform plasma in which Joule heating is balanced by radiative losses, with no radial heat transfer. In the present model, the state of the bulk plasma at the outer boundary is instead supplied from measurements or from an LTE model of the arc, and the radial heat loss of the column is included, which can be important, especially in the near-cathode region, where the temperatures are high [1, 32]. Gradients of the plasma properties, which drive ambipolar diffusion and heat fluxes, may exist at this boundary. The only requirement is that the plasma there be reasonably close to LTE. In this way, the model can directly receive the plasma state from an LTE model which facilitates the coupling. The near-electrode model and its numerical implementation are described in Section 2. The model is validated in Section 3 and applied to the near-cathode and near-anode layers of a high-intensity argon arc in Section 4.

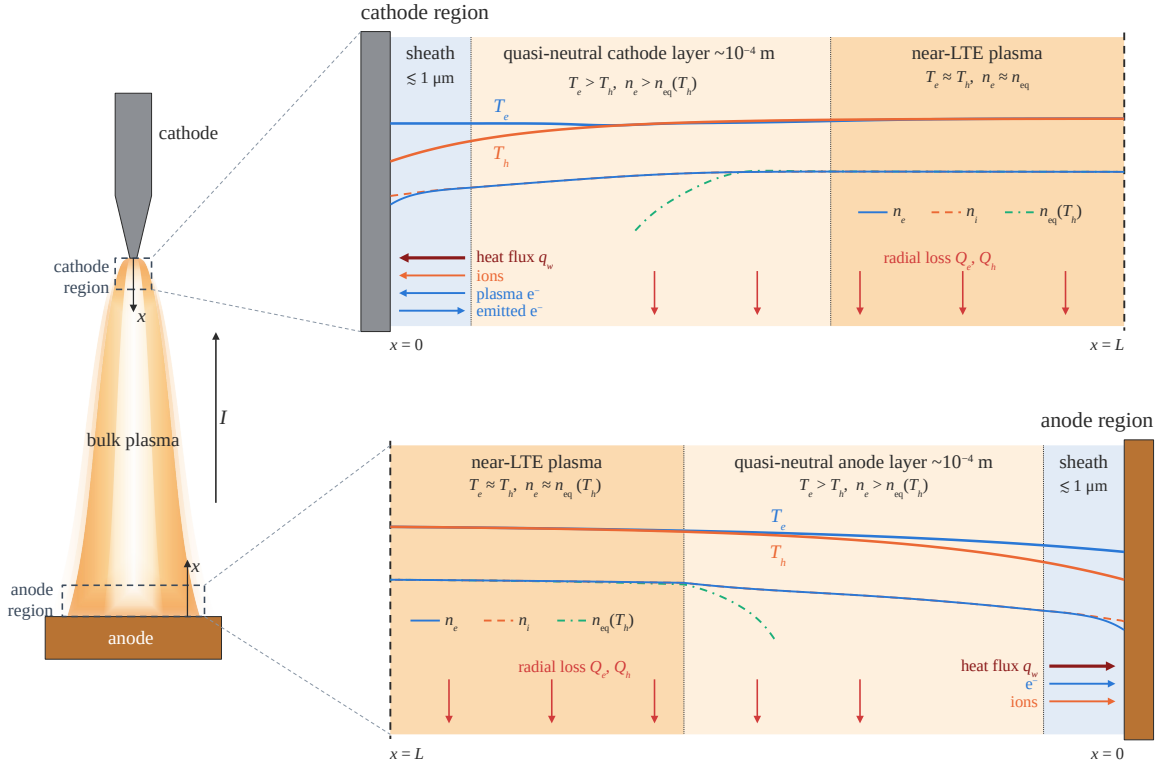


Figure 1. Schematic of the arc discharge and the modeled near-electrode regions.

2 Physical model and scale analysis

2.1 Spatial scale analysis

In the electrode regions the bulk plasma is perturbed by the presence of relatively cold and absorbing nature of electrode surfaces resulting in steep gradients in plasma properties. The bulk plasma therefore progressively deviates from the LTE conditions as the electrode is approached. We estimate the length scales over which these departures develop in a high-intensity atmospheric pressure arc considering LTE plasma adjacent to the electrodes at temperatures from 23 000 K, typical of the cathode region, down to 11 000 K, typical of the anode region [10, 33].

In the bulk plasma, the electron number density n_e is governed by ionization and recombination. Let n_n be the number density of argon atoms, $n_i = n_e$ that of the ions, and k_{ion} and k_{rec} the rate coefficients of electron-impact ionization and of three-body recombination. The electron density then evolves as $\dot{n}_e = k_{\text{ion}} n_e n_n - k_{\text{rec}} n_e^3$ (equation (2)). Linearizing this rate about equilibrium at fixed pressure p and temperature T, where Dalton's law gives $\delta n_n = -2\,\delta n_e$, yields the reaction time $\tau_{\text{rxn}} = 1/(2k_{\text{rec}} n_e^2 + 2k_{\text{ion}} n_e)$. During this time, the charged particles diffuse ambipolarly over the reaction length $\lambda_{\text{rxn}} = \sqrt{D_a \tau_{\text{rxn}}}$, where D_a is the ambipolar diffusion coefficient, equation (35). A plasma displaced from ionization equilibrium therefore travels a

Table 1. Characteristic length scales associated with the near-electrode regions of an high-intensity atmospheric pressure arc discharge.

T (K)	λ_{De} (m)	λ_e (m)	λ_i (m)	λ_{T_h} (m)	λ_{rxn} (m)	λ_T (m)
11 000	3.9×10^{-8}	9.2×10^{-7}	9.5×10^{-7}	1.8×10^{-6}	2.6×10^{-4}	3.4×10^{-4}
12 000	3.0×10^{-8}	6.0×10^{-7}	8.1×10^{-7}	2.1×10^{-6}	1.8×10^{-4}	2.3×10^{-4}
14 000	2.1×10^{-8}	3.8×10^{-7}	6.6×10^{-7}	3.2×10^{-6}	1.1×10^{-4}	1.4×10^{-4}
18 000	2.1×10^{-8}	4.5×10^{-7}	8.8×10^{-7}	5.9×10^{-6}	6.6×10^{-5}	1.7×10^{-4}
20 000	2.3×10^{-8}	5.7×10^{-7}	1.1×10^{-6}	6.9×10^{-6}	5.1×10^{-5}	2.2×10^{-4}
22 000	2.5×10^{-8}	7.2×10^{-7}	1.4×10^{-6}	7.8×10^{-6}	4.0×10^{-5}	2.8×10^{-4}
23 000	2.6×10^{-8}	8.1×10^{-7}	1.6×10^{-6}	8.2×10^{-6}	3.7×10^{-5}	3.1×10^{-4}

distance of order λ_{rxn} before reaching ionization equilibrium.

The electron and heavy-particle (ions and neutrals) temperatures in a plasma equilibrate through elastic collisions. In an elastic collision with a heavy particle, an electron loses only a fraction $2m_e/M$ of its energy, where m_e and M are the electron and argon masses. Its energy therefore relaxes only after about $M/(2m_e)$ collisions, during which it random-walks over a distance $\lambda_T = \lambda_{eh}/\sqrt{2m_e/M}$, which is the length scale of energy relaxation between the electrons and the heavy particles. Here $\lambda_{eh} = 1/(n_n\overline{Q}_{en} + n_i\overline{Q}_{ei})$ is the electron mean free path for heavy-particle collisions, and $\overline{Q}_{en}$ and $\overline{Q}_{ei}$ are the average momentum-transfer cross sections of electron–atom and electron–ion collisions over the electron energy distribution (Appendix A.1). Ions and atoms, by contrast, have nearly equal masses and exchange energy efficiently, so their temperatures equalize over $\lambda_{T_h} = 1/[(n_i + n_n)\overline{Q}_{in}]$, where $\overline{Q}_{in}$ is the average ion–atom momentum-transfer cross section over the ion energy distribution, dominated by resonant charge exchange (equation (34)).

The smallest scale is the electron Debye length $\lambda_{De} = \sqrt{\varepsilon_0 k_B T_e/(n_e e^2)}$, which is of order 10^{-8} m and governs the extent of charge separation. Here ε_0 is the vacuum permittivity, k_B the Boltzmann constant and e the elementary charge. These scales are calculated for the temperatures representative of the cathode and anode regions and are listed in table 1, together with the electron mean free path $\lambda_e = 1/(n_n\overline{Q}_{en} + n_i\overline{Q}_{ei} + n_e\overline{Q}_{ee})$ and the ion mean free path $\lambda_i = 1/(n_n\overline{Q}_{in} + n_i\overline{Q}_{ii})$, where $\overline{Q}_{ee} = \overline{Q}_{ei}$ and $\overline{Q}_{ii}$ is the average Coulomb cross section $\overline{Q}_{ei}$ evaluated at T_h.

The resulting hierarchy of length scales is $\lambda_{De} \ll \lambda_e \lesssim \lambda_i \lesssim \lambda_{T_h} \ll \lambda_{\mathrm{rxn}} \approx \lambda_T$. The largest scales, λ_{rxn} and λ_T, indicate that the electrode affects the plasma over a region of thickness of order 10^{-4} m. Other models and analyses [12, 24, 18, 25, 34, 14] arrive at similar estimates. Consistent with them, Lowke and Tanaka [35] found that departures from LTE are confined to within about 0.1 mm of the electrodes of a 200 A atmospheric-pressure argon arc, and suggested that the rest of the plasma can be modeled as being in LTE. Because λ_e and λ_i are small compared with this region, a continuum description with collision dominated transport is reasonable over the arc end of the near-electrode region, and because λ_{T_h} is small as well a single temperature T_h can be used to describe the heavy species. The validity of both assumptions close to the electrodes is examined a posteriori in Section 4.3.

2.2 Governing equations

This section describes the governing equations of the plasma in the electrode region. Externally imposed bulk flows, such as the cathode jet, are not modeled in the electrode region. The heavy gas is nevertheless not assumed to be at rest, since it can move during transients. The plasma is taken to consist of three species, namely electrons (e), singly charged argon ions (i), and argon atoms (n). Multiply charged ions are neglected, because the temperature in the anode region is too low for thier production, and in the cathode region even though they are present, they are shown to have a negligible effect [36]. The number densities of the three species obey

$$\frac{\partial n_e}{\partial t} + \nabla\cdot\mathbf{\Gamma}_e = S, \qquad \frac{\partial n_i}{\partial t} + \nabla\cdot\mathbf{\Gamma}_i = S, \qquad \frac{\partial n_n}{\partial t} + \nabla\cdot\mathbf{\Gamma}_n = -S, \tag{1}$$

where $\mathbf{\Gamma}_s = n_s\mathbf{u}_s$ is the flux of species s and S is the net ionization rate. The latter accounts for electron-impact ionization and its inverse, three-body recombination with an electron as the third particle,

$$S = k_{\mathrm{ion}}\, n_e n_n - k_{\mathrm{rec}}\, n_e^2 n_i. \tag{2}$$

The rate coefficients k_{ion} and k_{rec} have been evaluated in different ways by different authors [12, 24, 35, 37]. In this work we use those of Almeida et al. [12], taken from Benilov and Naidis [38]. In their model, ionization proceeds mainly stepwise, through the first excited level. The

escape of resonance radiation, which reduces the ionization rate, is accounted for when $n_e \lesssim 10^{21}$ m^{-3}, and k_{rec} follows from k_{ion} by detailed balance with Saha equilibrium at T_e.

The momentum equation for the plasma as a whole is $\rho\, D\mathbf{u}/Dt = \mathbf{F}$, where $\mathbf{F}$ is the net force on the plasma. For each species s, the corresponding equation is $m_s n_s\, D_s\mathbf{u}_s/Dt = \mathbf{F}_s$, where $\mathbf{F}_s$ is the net force on that species. We neglect the inertia of each species relative to the heavy gas, $D_s\mathbf{u}_s/Dt \simeq D\mathbf{u}/Dt$, so that every species shares the acceleration of the heavy gas and there is no relative acceleration between species. The inertia of species s is then $Y_s\mathbf{F}$, where $Y_s = m_s n_s/\rho$ is its mass fraction, and the species momentum equations reduce to the local force balances [39]

$$0 = -en_e\mathbf{E} - \nabla p_e - m_e n_e \bar{\nu}^M_{en}(\mathbf{u}_e - \mathbf{u}_n) - m_e n_e \bar{\nu}^M_{ei}(\mathbf{u}_e - \mathbf{u}_i) - C_e n_e k_B \nabla T_e - Y_e\mathbf{F}, \tag{3}$$

$$0 = +en_i\mathbf{E} - \nabla p_i - M n_i \bar{\nu}^M_{in}(\mathbf{u}_i - \mathbf{u}_n) - M n_i \bar{\nu}^M_{ie}(\mathbf{u}_i - \mathbf{u}_e) + C_i n_e k_B \nabla T_e - Y_i\mathbf{F}, \tag{4}$$

$$0 = -\nabla p_n - M n_n \bar{\nu}^M_{ni}(\mathbf{u}_n - \mathbf{u}_i) - M n_n \bar{\nu}^M_{ne}(\mathbf{u}_n - \mathbf{u}_e) + C_n n_e k_B \nabla T_e - Y_n\mathbf{F}. \tag{5}$$

The frequencies $\bar{\nu}^M_{en}$, $\bar{\nu}^M_{ei}$, $\bar{\nu}^M_{in}$, $\bar{\nu}^M_{ie}$, $\bar{\nu}^M_{ni}$, and $\bar{\nu}^M_{ne}$ are *averaged momentum-transfer collision frequencies* within the collision pairs, the superscript m has been used to distinguish them from the *averaged energy-transfer collision frequencies* $\bar{\nu}^E_{ab}$, C_e is the electron thermal-force coefficient, and C_i and C_n are its reactions on the ions and atoms. Their calculation is described in Appendix A. Momentum conservation in each collision pair relates the frequencies, and the thermal forces sum to zero,

$$m_e n_e \bar{\nu}^M_{en} = M n_n \bar{\nu}^M_{ne}, \qquad m_e n_e \bar{\nu}^M_{ei} = M n_i \bar{\nu}^M_{ie}, \qquad n_i \bar{\nu}^M_{in} = n_n \bar{\nu}^M_{ni}, \qquad C_i + C_n = C_e. \tag{6}$$

Adding equations (3)–(5) and using equation (6) gives the total momentum equation,

$$\rho\, D\mathbf{u}/Dt = e(n_i - n_e)\mathbf{E} - \nabla p. \tag{7}$$

In practice, equations (3), (4) and (7) are solved instead of (5), and the neutral flux is obtained from the definition of the heavy-gas velocity $\mathbf{u}$ as the mass-weighted average of the ion and neutral velocities,

$$\mathbf{\Gamma}_n = (n_i + n_n)\mathbf{u} - n_i\mathbf{u}_i. \tag{8}$$

Solving the electron balance (3) for the electron flux, with $\bar{\nu}^M_e = \bar{\nu}^M_{en} + \bar{\nu}^M_{ei}$ the total electron momentum-transfer frequency and $\nabla p_e = k_B T_e \nabla n_e + n_e k_B \nabla T_e$, gives

$$\mathbf{\Gamma}_e = -\frac{en_e}{m_e\bar{\nu}^M_e}\mathbf{E} - \frac{k_B T_e}{m_e\bar{\nu}^M_e}\nabla n_e - (1 + C_e)\,\frac{n_e k_B}{m_e\bar{\nu}^M_e}\nabla T_e + \frac{n_e}{\bar{\nu}^M_e}\left(\bar{\nu}^M_{en}\mathbf{u}_n + \bar{\nu}^M_{ei}\mathbf{u}_i\right), \tag{9}$$

where the negligible term $Y_e\mathbf{F}$ has been dropped. The first two terms are the drift and diffusion fluxes. Their coefficients, $e/(m_e\bar{\nu}^M_e)$ and $k_B T_e/(m_e\bar{\nu}^M_e)$, are the magnitudes of the electron mobility and diffusivity. Because both are set by the same frequency, they satisfy the Einstein relation identically. The third term combines the temperature-gradient part of the electron pressure force with the thermal force. The last term is the collisional drag exerted on the electrons by the moving heavy species.

The ion flux follows from the ion balance (4) in a similar way,

$$\mathbf{\Gamma}_i = \frac{en_i}{M\bar{\nu}^M_i}\mathbf{E} - \frac{k_B T_h}{M\bar{\nu}^M_i}\nabla n_i - \frac{n_i k_B}{M\bar{\nu}^M_i}\nabla T_h + C_i\,\frac{n_e k_B}{M\bar{\nu}^M_i}\nabla T_e - \frac{n_i}{n_h}\frac{\mathbf{F}}{M\bar{\nu}^M_i} + \frac{n_i}{\bar{\nu}^M_i}\left(\bar{\nu}^M_{in}\mathbf{u}_n + \bar{\nu}^M_{ie}\mathbf{u}_e\right). \tag{10}$$

The first three terms are ion drift, diffusion, and diffusion driven by the heavy-particle temperature gradient, with mobility $e/(M\bar{\nu}^M_i)$ and diffusivity $k_B T_h/(M\bar{\nu}^M_i)$. The fourth term is the reaction of the ions to the electron thermal force. The fifth term is the ions' share, $Y_i\mathbf{F}$, of the net force on the plasma, which accelerates them together with the rest of the heavy gas. The last term is the collisional drag exerted on the ions by the atoms and electrons. The velocities in the drag terms of equations (9) and (10) are those of the other species, so the two fluxes are coupled. If the drag, thermal-force and mixture-acceleration terms are removed ($C_e = C_i = 0$, $Y_i\mathbf{F} = 0$, and $\mathbf{u}_s = \mathbf{u} = 0$ in the drag terms), equations (9) and (10) reduce to the conventional drift diffusion equations. Unlike the classical fluxes, however, they retain the full pressure-gradient force of each species in a non-isothermal plasma.

The electric field is obtained from Poisson's equation,

$$\nabla\cdot(\varepsilon_0\mathbf{E}) = e(n_i - n_e), \qquad \mathbf{E} = -\nabla\phi, \tag{11}$$

which is solved over the entire computational domain, so that the transition from the quasi-neutral bulk plasma to the space-charge region is captured without imposing the sheath boundaries a priori.

The electron thermal energy density $\mathcal{E} = \frac{3}{2} n_e k_B T_e$ obeys

$$\frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot \mathbf{h}_e = -e\,\mathbf{\Gamma}_e \cdot \mathbf{E} - \frac{3}{2} n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^{E} - \varepsilon_{\text{ion}} S - w_{\text{rad}} - Q_e, \tag{12}$$

with the electron energy flux

$$\mathbf{h}_e = \frac{5}{2} k_B T_e\,\mathbf{\Gamma}_e - \kappa_e \nabla T_e + k_B T_e\, n_e \Big[C_i(\mathbf{u}_e - \mathbf{u}_i) + C_n(\mathbf{u}_e - \mathbf{u}_n) \Big]. \tag{13}$$

The terms on the right-hand side of equation (12) are Joule heating by the field, elastic energy transfer to the heavy species, the ionization-energy sink, the net radiative loss, and the radial heat loss of the electrons. In the elastic term, $\bar{\nu}_{eh}^{E} = \bar{\nu}_{en}^{E} + \bar{\nu}_{ei}^{E}$, where the superscript E denotes the *averaged energy-transfer collision frequencies* (Appendix A). The ionization sink is the net ionization rate multiplied by $\varepsilon_{\text{ion}} = 15.76$ eV, the first ionization potential of argon. The energy flux (13) is the sum of the enthalpy flux at $\frac{5}{2} k_B T_e$ per electron, Fourier conduction with the zero-current electron thermal conductivity κ_e, and a thermal-diffusion heat flux [39]. The net radiative loss w_{rad} is that of Almeida et al. [12], taken from Benilov and Naidis [38].

A one-dimensional domain has no lateral boundary, so the Joule heat generated away from the electrode can leave only by radiation and by conduction through the two ends of the domain. In a real arc, by contrast, the bulk plasma can lose heat radially, by conduction [32]. We therefore represent the radial conduction of a column of radius R by a volumetric loss for each energy channel [32],

$$Q_e = \Lambda\,(\kappa_e + \kappa_R)\,(T_e - T_{\text{edge}}), \qquad Q_h = \Lambda\,\kappa_h\,(T_h - T_{\text{edge}}), \qquad \Lambda = \frac{4}{R^2}, \tag{14}$$

where T_{edge} is the temperature at the edge of the column. Ion pairs also diffuse radially out of the hot core and recombine near the cold edge, releasing their ionization energy there. This heat loss is represented by the reactive thermal conductivity κ_R (Appendix A.5), which is assigned to the electron channel [17]. The radius R is a parameter of the model. We take it as the radius of a uniform column that carries the arc current I at the applied current density, $I = \pi R^2 J_{\text{app}}$, so that $R = \sqrt{I/(\pi J_{\text{app}})}$. Since the loss scales as $\kappa T/R^2$, it is most important where the plasma is hot and the attachment is small, as in the cathode spot. The radius therefore differs between the cathode and the anode, and it changes with the current density (Section 4). Closer to the electrodes, the axial gradients develop over 10^{-6}–10^{-4} m, whereas R is of order 10^{-3} m, so the radial losses are comparatively negligible there.

The common heavy-particle temperature is governed by

$$n_h c_v \left(\frac{\partial T_h}{\partial t} + \mathbf{u} \cdot \nabla T_h \right) = \nabla \cdot (\kappa_h \nabla T_h) + \frac{3}{2} n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^{E} + Q_{in} - Q_h - n_h k_B T_h\,\nabla \cdot \mathbf{u}, \tag{15}$$

where $c_v = \frac{3}{2} k_B$ is the translational heat capacity per heavy particle and κ_h is the heavy-particle thermal conductivity (Appendix A). The elastic-exchange term is the same as in the electron energy equation (12), but with the opposite sign. The ionization energy and the radiative loss are assigned entirely to the electrons and do not appear here. The term Q_h is the radial heat loss of the heavy particles, equation (14), and the ion–neutral frictional heating is

$$Q_{in} = M n_i\,\bar{\nu}_{in}^{M}\,|\mathbf{u}_i - \mathbf{u}_n|^2, \tag{16}$$

which is positive definite and reduces to the Joule dissipation of the ion drift when the neutral gas is at rest. The last term in equation (15) is the compression work of the heavy gas.

The velocity of the heavy gas as a whole is governed by an inviscid single-fluid momentum equation,

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + e(n_i - n_e)\mathbf{E}, \qquad p = n_e k_B T_e + n_h k_B T_h. \tag{17}$$

We therefore solve for following qunatities, $\{n_e, n_i, n_n, \mathcal{E}, T_h, \rho u, \phi\}$. They are governed by the continuity equations (1) with the fluxes (9), (10) and (8), Poisson's equation (11), the electron and heavy-particle energy equations (12) and (15), and the heavy-gas momentum equation (17).

2.3 Boundary conditions

The system is closed by conditions at the two ends of the domain, at the electrode surface, $x = 0$, and at the bulk-plasma boundary $x = L$.

Electrode surface, $x = 0$. Electrons reaching the surface are absorbed, and they are assumed to have a Maxwellian distribution at their local temperature T_e. At the cathode, additionally the thermionic emission flux Γ_{em} of electrons with a Maxwellian distribution at the wall temperature T_{wall} is included. With $\hat{\mathbf{n}}$ the unit vector normal to the surface and pointing into the electrode, the electron flux into the surface is therefore given by,

$$\boldsymbol{\Gamma}_e \cdot \hat{\mathbf{n}} = \tfrac{1}{4} n_e \bar{c}_e - \Gamma_{\mathrm{em}}, \qquad \bar{c}_e = \sqrt{\frac{8 k_B T_e}{\pi m_e}}. \tag{18}$$

The emitted flux is given by the Richardson–Dushman law with Schottky lowering of the work function W by the field at the surface,

$$\Gamma_{\mathrm{em}} = \frac{A_R T_{\mathrm{wall}}^2}{e} \exp\!\left(-\frac{W - \Delta W}{k_B T_{\mathrm{wall}}}\right), \qquad \Delta W = \sqrt{\frac{e^3 E_w}{4\pi\varepsilon_0}}, \qquad E_w = \max(\mathbf{E} \cdot \hat{\mathbf{n}}, 0), \tag{19}$$

where A_R is the Richardson constant and W the work function. Each absorbed electron removes a mean energy $2k_B T_e$ from the plasma, and each emitted electron brings in $2k_B T_{\mathrm{wall}}$, so the boundary condition for the electron energy equation (12) is

$$\mathbf{h}_e \cdot \hat{\mathbf{n}} = 2 k_B T_e \, \tfrac{1}{4} n_e \bar{c}_e - 2 k_B T_{\mathrm{wall}} \, \Gamma_{\mathrm{em}}. \tag{20}$$

Ions reaching the surface are assumed to recombine there and to return to the plasma as atoms. The ion flux into the surface is the sum of a thermal-effusion flux and of the directed part of the ion flux when it points into the wall. With $\boldsymbol{\Gamma}_i^d = \boldsymbol{\Gamma}_i + [k_B T_h/(M \bar{\nu}_i^M)] \nabla n_i$, the ion flux (10) without its density-diffusion term,

$$\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}} = \tfrac{1}{4} n_i \bar{c}_i + \max\!\left(\boldsymbol{\Gamma}_i^d \cdot \hat{\mathbf{n}}, 0\right), \qquad \bar{c}_i = \sqrt{\frac{8 k_B T_{\mathrm{wall}}}{\pi M}}. \tag{21}$$

Where the sheath field drives the ions into the electrode, the directed term dominates. Elsewhere, the thermal-effusion flux ensures that ions reaching the wall are absorbed even where transport alone does not carry them into it. The ion current density into the electrode is $j_i = e \boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}}$. Within the layer, j_i is defined in the same way, with $\hat{\mathbf{n}}$ the unit vector towards the electrode, so that $j_i > 0$ wherever the ions flow towards the electrode, at the cathode and at the anode alike.

The wall is impermeable to the heavy gas, $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, which is the boundary condition for the heavy-gas momentum equation (17). Equation (8) then gives

$$\boldsymbol{\Gamma}_n \cdot \hat{\mathbf{n}} = -\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}}, \tag{22}$$

so that every ion neutralized at the surface re-enters the plasma as an atom. Secondary electron emission by ion impact is not included.

The heavy-particle temperature at the surface equals the electrode temperature, $T_h(0) = T_{\mathrm{wall}}$. At the cathode, T_{wall} is obtained by balancing the heat flux q_w delivered by the plasma against steady conduction through the cathode body,

$$q_w = \frac{k_w}{L_w} \left(T_{\mathrm{wall}} - T_0\right), \tag{23}$$

where k_w is the thermal conductivity of the cathode, L_w its length, and T_0 the temperature of its base. This simple heat balance is intended to capture the trend of the cathode temperature with the current. The anode is usually water-cooled, and its surface temperature is prescribed.

The heat flux delivered at the electrode surface can be calculated as,

$$q_w = (2 k_B T_e + W) \, \tfrac{1}{4} n_e \bar{c}_e - (2 k_B T_{\mathrm{wall}} + W) \, \Gamma_{\mathrm{em}} + (\varepsilon_{\mathrm{ion}} - W) \, \boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}} - \kappa_h \nabla T_h \cdot \hat{\mathbf{n}}. \tag{24}$$

The first term is the energy delivered by the absorbed electrons, which is their kinetic energy plus the work function released on entering the metal. The second is emission cooling, since each emitted electron removes the work function and its kinetic energy. The third is the energy released when an ion recombines at the surface, $\varepsilon_{\mathrm{ion}}$, less the work function spent to extract the neutralizing electron. The last is heat conduction from the heavy gas. At the anode, $\Gamma_{\mathrm{em}} = 0$ and the emission term vanishes. The electrodes are given a zero reference potential.

Bulk-plasma edge, $x = L$. The electron and ion densities and both temperatures are set to Dirichlet values from the adjacent LTE bulk plasma, $n_e = n_e^{\mathrm{bulk}}$, $n_i = n_i^{\mathrm{bulk}}$ and $T_e = T_h = T_{\mathrm{bulk}}$,

the last of which is the boundary condition for both energy equations. The neutral density follows from Dalton's law at the background pressure p_0,

$$n_n = \frac{p_0 - n_e k_B T_e}{k_B T_h} - n_i. \tag{25}$$

This edge is open to the heavy gas. In the heavy-gas momentum equation (17), the state outside the boundary is the bulk state at the pressure p_0, and the heavy-gas velocity has a zero normal gradient. The arc current is imposed through the electric field in Poisson's equation (11). The total current density at the boundary, conduction plus displacement, is held equal to the applied current density J_{app},

$$e\,(\mathbf{\Gamma}_i - \mathbf{\Gamma}_e) + \varepsilon_0 \frac{\partial E}{\partial t} = J_{\mathrm{app}}, \tag{26}$$

which is integrated in time as an evolution equation for the boundary field. The ion fraction of the current is given as j_i/j, with $j = |J_{\mathrm{app}}|$.

2.4 Numerical implementation

The model is solved with a finite-volume code built on the block-structured adaptive mesh refinement framework AMReX [40]. The domain $0 \le x \le L$ is divided into cells, and the cell averages of n_e, n_i, n_n, $\mathcal{E}$, T_h, ρu and ϕ are the discrete unknowns. Every conservation law is written in flux form, with the fluxes evaluated at the cell faces. The species fluxes (9) and (10) are split into a density-diffusion part, $-D\,\partial n/\partial x$, and a non-diffusive part, nv, where the velocity v collects the drift in the field, the temperature-gradient and thermal-force terms, the mixture force and the collisional drag. In the sheath, the drift and diffusion fluxes of the electrons nearly cancel, and their sum is about two orders of magnitude smaller than either Section (4.3). The face flux is therefore the exponentially fitted (Scharfetter–Gummel) flux [41],

$$\Gamma_f = \frac{D_f}{\Delta x}\Big[B(-\psi)\,n_{\mathrm{L}} - B(\psi)\,n_{\mathrm{R}}\Big], \qquad \psi = \frac{v_f\,\Delta x}{D_f}, \qquad B(\psi) = \frac{\psi}{e^{\psi} - 1}, \tag{27}$$

where v_f and D_f are the non-diffusive velocity v and the diffusion coefficient D of the species evaluated at the face, L and R denote the cells on either side of the face, and ψ is the cell Péclet number. Equation (27) is exact for the exponential profile of a Boltzmann balance, and it reduces to central differencing as $\psi \to 0$ and to upwinding as $|\psi| \to \infty$. The enthalpy part of the electron energy flux (13) is computed from this discrete electron flux, $\frac{5}{2}k_B T_e\,\Gamma_{e,f}$, so that each electron crossing a face carries exactly its own enthalpy. Heat conduction in both energy equations and Poisson's equation (11) are discretized with central differences, and the face field $E_f = -(\phi_{\mathrm{R}} - \phi_{\mathrm{L}})/\Delta x$ is used in every face flux and in the Joule heating. The heavy-gas momentum equation (17) is discretized with a Roe-type flux [42]. At the electrode, the wall fluxes (18)–(22) are imposed as the face fluxes, evaluated with the densities and temperatures of the cell adjacent to the wall. At $x = L$, the Dirichlet values are imposed on the boundary face.

All cases in this study use a uniform base mesh of 512 cells and four levels of refinement towards the electrode, each with a refinement ratio of 4. The refined regions are nested layers adjacent to the electrode, 1.0×10^{-4}, 2.0×10^{-6}, 7.3×10^{-7} and 2.7×10^{-7} m deep, with cell sizes of 244, 61, 15.3 and 3.8 nm, so that the mesh has 944 cells in total. These sizes are chosen to resolve the length scales estimated in Section 2.1, down to a fraction of the Debye length, at both the cathode and the anode. The levels are coupled conservatively. Starting from a uniform initial state at the bulk-plasma conditions or from a neighboring steady state, the equations are advanced in time until a steady state is reached. All equations, including Poisson's equation and the galvanostatic condition (26), are solved together at each time step. A run is considered steady when, for each of ϕ, n_e, n_i, $\mathcal{E}$ and T_h, the relative change over a time step, $\|\mathbf{U}^{n+1} - \mathbf{U}^{n}\|_2 / \|\mathbf{U}^{n+1}\|_2$, scaled to 1 μs of simulated time, stays below a tolerance of at most 10^{-5} on several consecutive checks.

3 Validation

The model is validated against two published unified models of the near-electrode layers of atmospheric-pressure argon arcs, the near-cathode layer computed by Almeida et al. [12] and the near-anode layer computed by Semenov et al. [14]. Both works solve the multi-fluid equations for electrons, ions and atoms, with separate electron and heavy-particle temperatures, coupled to Poisson's equation. Their geometries differ, however. Semenov et al. use a planar one-dimensional model, as the present one does, so the current density is the same at every x. Almeida et al. instead solve their equations in spherical symmetry about a hemispherical cathode of radius

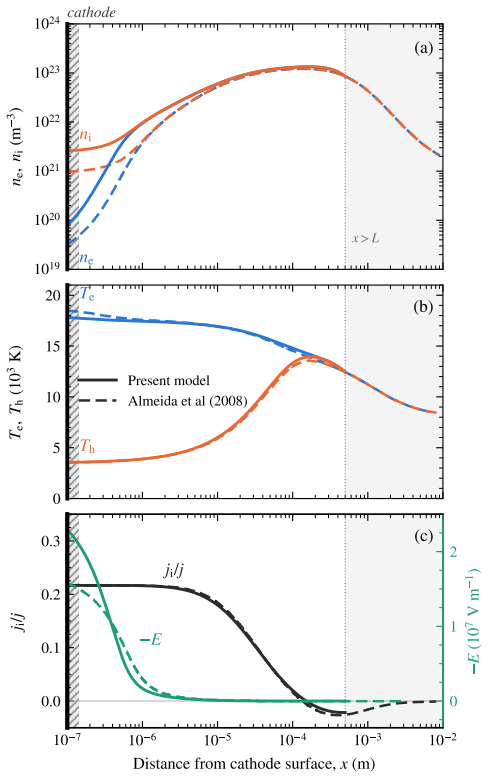


Figure 2. Near-cathode layer ($T_w = 3500$ K, $j_c = 10^7$ A m^{-2}): present model (solid) and Almeida et al. [12] (dashed). (a) Number densities. (b) Temperatures. (c) j_i/j (left axis) and $-E$ (right axis). The shaded region, $x > L$, is outside the domain.

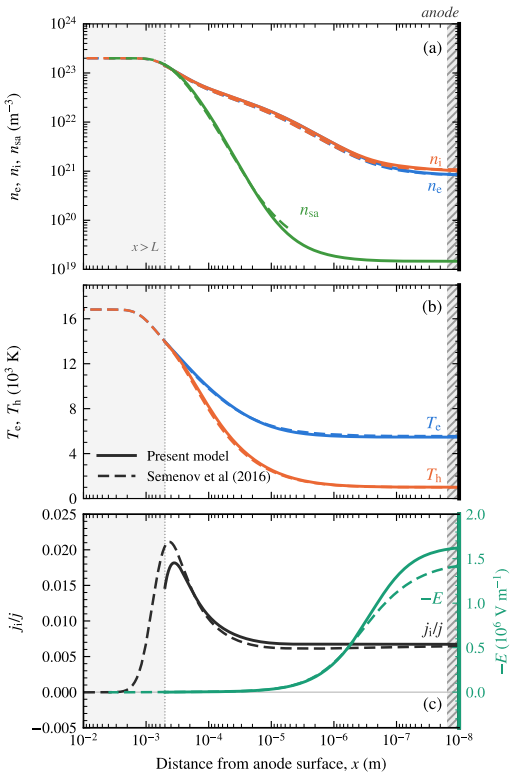


Figure 3. Near-anode layer ($T_s = 1000$ K, $j = 1.5 \times 10^7$ A m^{-2}): present model (solid) and Semenov et al. [14] (dashed). (a) Number densities and the Saha density n_{sa}. (b) Temperatures. (c) j_i/j (left axis), and $-E$, towards the anode (right axis). The shaded region, $x > L$, is outside the domain.

$r_c = 1$ mm, so their current density decreases with distance as $j_c\, r_c^2/(r_c + x)^2$, whereas the present model keeps $j = j_c$ throughout. At the outer boundary of their domains, 1 cm from the electrode, both assume that the plasma is uniform and in LTE, with Joule heating balanced by radiation alone. Their solutions show, however, that the plasma is already close to LTE much nearer the electrode. At $x = 5 \times 10^{-4}$ m, T_e and T_h differ by less than 0.3 % in both published solutions, and the charged-particle density is within 2 % of the Saha density, although the plasma there is not yet uniform. We therefore restrict the present domain to $L = 5 \times 10^{-4}$ m and take the boundary values at $x = L$ from the published solutions.

The conditions of the published calculations were reproduced as closely as possible. The electrode surface temperature was prescribed at the value used in each work, rather than obtained from the heat balance (23), and the radial loss (14) was switched off, $\Lambda = 0$, since neither model includes it. The domain was discretized as described in Section 2.4. Throughout the space-charge sheath, the region where $|n_i - n_e|/n_i > 1\,\%$,the cells are smaller than the local electron Debye length, by a factor of at least 1.4 at the cathode and 6 at the anode. The equations were integrated in time at a fixed current density to a steady state.

3.1 Near-cathode layer

Almeida et al. [12] computed the near-cathode layer of an atmospheric-pressure argon arc for three current densities and concluded that only one of these cases is realizable in practice, namely a tungsten cathode with a work function of 4.55 eV, thermionic emission (19) included, a surface temperature of 3500 K and a current density of 10^7 A m^{-2}. We therefore model this case, and the two solutions are compared in figure 2. The temperature profiles agree closely across the whole layer (figure 2(b)). The electron temperature stays at 17.3–17.8 kK through the sheath and the inner part of the ionization layer, $x \lesssim 3 \times 10^{-6}$ m, and decreases towards the bulk plasma, while the heavy-particle temperature rises from the wall value to a maximum of 13.9 kK at $x \approx 1.7 \times 10^{-4}$ m, compared with 13.6 kK in the solution of Almeida et al. From $x \approx 3 \times 10^{-6}$ m outwards, both temperatures agree with theirs within 0.4 kK (3 %).

The ion fraction of the current, figure 2(c), is reproduced over the whole layer. At the cathode

surface the present model gives $j_i/j = 0.217$, compared with 0.216 reported by Almeida et al. The ion fraction stays at this value through the sheath, falls to zero at $x \approx 1.5 \times 10^{-4}$ m and becomes slightly negative further out, where the ion flux is directed away from the cathode. Almeida et al. place the sheath edge where the charge separation reaches 1 %. With the same definition, the present model gives a sheath voltage of 10.0 V at $x_s = 1.1$ μm, compared with 10.2 V at $x_s = 1.5$ μm in their solution. In the region, $10^{-5} \lesssim x \lesssim 3 \times 10^{-4}$ m, the charged-particle densities of the present model are 7–15 % higher than theirs (figure 2(a)). The largest differences are in the space-charge sheath, $x \lesssim 10^{-6}$ m. At $x = 10^{-7}$ m, the electron and ion densities of the present model are higher by a factor of about 2.5, and the field strength is 2.3×10^7 V m^{-1}, compared with 1.6×10^7 V m^{-1}.

3.2 Near-anode layer

For anode we consider the case corresponding to figure 3(b) of Semenov et al. [14] which represents a non-emitting anode $\Gamma_{em} = 0$, with a surface temperature of 1000 K and a current density of 1.5×10^7 A m^{-2}. The comparison is shown in figure 3. The electron and ion densities agree with those of Semenov et al. within 15 % across the layer (figure 3(a)). Following Semenov et al., the figure also shows the Saha density n_{sa}, the charged-particle density that the local T_e and T_h would give in ionization equilibrium with quasi-neutrality at the pressure p_0. Within 10 μm of the anode, the actual densities exceed n_{sa} by about two orders of magnitude in both models. The anode region is therefore far from ionization equilibrium, and the charged particles are supplied by diffusion from the hotter plasma rather than produced locally. The temperature profiles agree closely (figure 3(b)). At the anode surface, the electron temperature is 5.46 kK, compared with 5.59 kK reported by Semenov et al., and the heavy-particle temperatures differ by less than 0.3 kK across the layer. The electric field near the anode, figure 3(c), is directed towards the anode, so the anode lies at a lower potential than the adjacent plasma and the electrons are retarded. The field strength at the surface is 1.65×10^6 V m^{-1}, compared with 1.42×10^6 V m^{-1} in the solution of Semenov et al. The ions flow into the anode, $j_i > 0$. At the surface, the present model gives $j_i/j = 0.0067$, compared with their 0.0065. With the same definition of the sheath edge as at the cathode, the present model gives sheath width $x_s = 0.69$ μm and sheath voltage $\phi_s = 0.58$ V, compared with $x_s = 0.71$ μm and $\phi_s = 0.54$ V obatined by Semenov et al. The main difference is in the maximum of j_i/j, which is 0.018 at $x = 3.5 \times 10^{-4}$ m in the present model and 0.021 at $x = 4.3 \times 10^{-4}$ m in the solution of Semenov et al. This difference reflects the differences between the models. Imposing their state at $x = L$ fixes the values of the densities and temperatures there, but not their gradients, which follow from each model's governing equations and on which j_i/j depends.

Overall, the present model reproduces the structure of both the near-cathode and the near-anode layer, including the temperatures, the current partition between ions and electrons, the sign and magnitude of the near-electrode field, the sheath voltages (within 2 % at the cathode and 8 % at the anode) and the departure from ionization equilibrium. Apart from the charged-particle densities in the cathode sheath and the maximum of j_i/j near the outer boundary of the anode calculation, the temperatures agree within 3–5 % and the charged-particle densities within 15 %. The differences most likely arise from the different descriptions of the transport properties (Appendix A) and of the ion boundary condition (21).

4 Results

The model is now applied to the cathode and anode regions of a high-intensity argon arc at atmospheric pressure. Experiments and LTE models of such arcs give the state of the plasma near the electrodes. On the basis of the scale estimates of Section 2.1 and of previous studies [12, 24, 34, 14], the plasma is assumed to be in LTE at a distance of 5×10^{-4} m from the electrodes. At this distance, $x = L$, we take $T_{bulk} = 21\,000$ K and $J_{app} = 10^8$ A m^{-2} for the cathode, and $T_{bulk} = 15\,000$ K and $J_{app} = 4 \times 10^6$ A m^{-2} for the anode which are typical values on the axis of a 200 amps arc [10, 33, 8, 37, 11]. All densities and temperatures at $x = L$ are supplied as Dirichlet values for both cases (Section 2.3). Unlike in the validation cases of Section 3, the radial losses (14) are included, with $T_{edge} = 1000$ K. The column radius is that of a column carrying the arc current $I = 200$ A at the applied current density, $R = \sqrt{I/(\pi J_{app})}$ (Section 2.2), which gives $R = 0.80$ mm at the cathode and $R = 3.99$ mm at the anode. The small cathode radius reflects the attachment of the arc in a hot spot [10, 37], and the larger anode radius its wider attachment [10, 8, 11]. Bulk-plasma at the cathode is almost fully ionized, and the reactive thermal conductivity $\kappa_R = 0.03$ W m^{-1} K^{-1} is negligible compared with electron thermal conductivity $\kappa_e = 2.6$ W m^{-1} K^{-1}. At anode however, $\kappa_R = 0.88$ W m^{-1} K^{-1}, is comparable to $\kappa_e = 1.27$ and exceeds $\kappa_h = 0.12$ W m^{-1} K^{-1}.

The LTE models and experiments from which the boundary conditions are obtained uses copper as anode and thoriated tungsten as cathode, and the same has been used in this work. The anode has a work function of 4.65 eV and a prescribed, uniform surface temperature $T_{\text{wall}} = 1000$ K. The cathode surface temperature follows from the heat balance (23), with the uniform properties $k_w = 100\ \text{W m}^{-1}\,\text{K}^{-1}$, $L_w = 1$ cm and $T_0 = 1000$ K, and the emission parameters $A_R = 3 \times 10^4\ \text{A m}^{-2}\,\text{K}^{-2}$ and $W = 2.63$ eV. The computational mesh as described in Section 2.4 is employed. Table 2 summarizes the supplied and computed parameters for the cathode and anode cases.

Table 2. Conditions (top) and computed electrode quantities (bottom). The sheath edge x_s is the distance from the electrode at which $|n_i - n_e|/n_i$ falls to 1 %, and ϕ_s is the potential there. The potential at the bulk-plasma boundary is $\phi(L)$, and both potentials are relative to the electrode. Fluxes, the ion current density j_i and the field are positive towards the electrode.

	Cathode	Anode
Current density J_{app} (A m^{-2})	10^8	4×10^6
Bulk temperature T_{bulk} (K)	21 000	15 000
Bulk electron density n_e^{bulk} (m^{-3})	1.7×10^{23}	1.8×10^{23}
Column radius R (mm)	0.80	3.99
Thermionic emission	on	off
Surface temperature T_{wall} (K)	3714	1000 (prescribed)
T_e at the surface (K)	17 100	6100
n_e, n_i at the surface (m^{-3})	2.6×10^{21}, 6.6×10^{21}	2.1×10^{20}, 6.6×10^{20}
Sheath edge, x_s (m)	5.6×10^{-7}	1.0×10^{-6}
ϕ_s, $\phi(L)$ (V)	3.1, 11.7	1.39, 5.22
Field at the surface $\mathbf{E} \cdot \hat{\mathbf{n}}$ (V m^{-1})	1.6×10^7	3.8×10^6
j_i/j at the surface	0.048	0.027
Absorbed electron flux $\frac{1}{4} n_e \bar{c}_e$ ($\text{m}^{-2}\,\text{s}^{-1}$)	5.2×10^{26}	2.6×10^{25}
Emitted electron flux Γ_{em} ($\text{m}^{-2}\,\text{s}^{-1}$)	1.11×10^{27}	–
Ion flux $\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}}$ ($\text{m}^{-2}\,\text{s}^{-1}$)	3.0×10^{25}	6.8×10^{23}
Heat flux q_w (MW m^{-2})	27.2	35.7

4.1 Structure of the near-electrode layers

Cathode. A space-charge sheath 0.56 μm thick forms at the surface. In it, n_i exceeds n_e, by up to a factor of 2.6 at the surface (figure 4(a)), and across it the potential rises by 3.1 V. The field at the surface is $1.6 \times 10^7\ \text{V m}^{-1}$, and it falls by an order of magnitude across the sheath (figure 4(e)). Beyond the sheath, the potential keeps rising through the quasi-neutral ionization layer, to 7.9 V at $x = 10^{-4}$ m. Further out, the plasma approaches the bulk state, and the remaining rise to 11.7 V at $x = L$ is mostly the ohmic drop of the column, whose field is $7.5 \times 10^3\ \text{V m}^{-1}$. The electron temperature varies little across the whole layer, 17.1–22.6 kK. It is highest, 22.6 kK, at $x \approx 0.36$ mm, above T_{bulk}, and falls to T_{bulk} at $x = L$. The heavy-particle temperature rises from the surface value of 3.7 kK and approaches T_e only at $x \approx 10^{-4}$ m (figure 4(c)). Over the same distance, the plasma becomes almost fully ionized. Close to the cathode, the charged-particle density is far above the equilibrium density $n_{\text{eq}}(T_h)$ that the reactions would give at the local heavy-particle temperature (figure 4(a)), by a factor of 360 at 10 μm. The two densities come within a factor of 2 of each other beyond 32 μm, and within 1 % only beyond 0.19 mm, where T_h has approached T_e. The emitted electron flux, $1.11 \times 10^{27}\ \text{m}^{-2}\,\text{s}^{-1}$, corresponds to a current density of $1.79 \times 10^8\ \text{A m}^{-2}$, 1.79 times the arc current density. It is partly offset by the plasma electrons that reach the surface against the sheath field, $5.2 \times 10^{26}\ \text{m}^{-2}\,\text{s}^{-1}$ (table 2), and the ions carry 4.8 % of the current at the surface (figure 4(e)).

Anode. The space-charge sheath is thin. The charge separation exceeds 1 % only within 1.0 μm of the surface, across which the potential changes by 1.39 V. In this sheath the ions are in excess, and at the surface n_i exceeds n_e by a factor of 3.1 (figure 4(b)). The anode lies 5.22 V below the bulk-plasma boundary, and most of this drop is distributed over the quasi-neutral layer, with 2.38 V within 10 μm and 3.44 V within 100 μm. The field is directed towards the anode, $3.8 \times 10^6\ \text{V m}^{-1}$ at the surface (figure 4(f)), and retards the electrons, which reach the surface by diffusion down the steep density gradient. Across the layer, the charged-particle density falls by a factor of 870, from 1.8×10^{23} to $2.1 \times 10^{20}\ \text{m}^{-3}$. The charged-particle densities exceed the equilibrium density $n_{\text{eq}}(T_h)$ by a factor of 33 at 100 μm, 2.3 at 200 μm and 1.2 at 300 μm (figure 4(b)), while closer to the anode $n_{\text{eq}}(T_h)$ is negligible. The anode layer is thus over-populated with charged particles, which recombine there, and n_e comes within 10 % of $n_{\text{eq}}(T_h)$ only within 0.15 mm of $x = L$. The electron and heavy-particle temperatures decrease together towards the anode, but T_h falls faster, to the prescribed surface temperature of 1 kK, so the two separate progressively, by 1.3 kK at

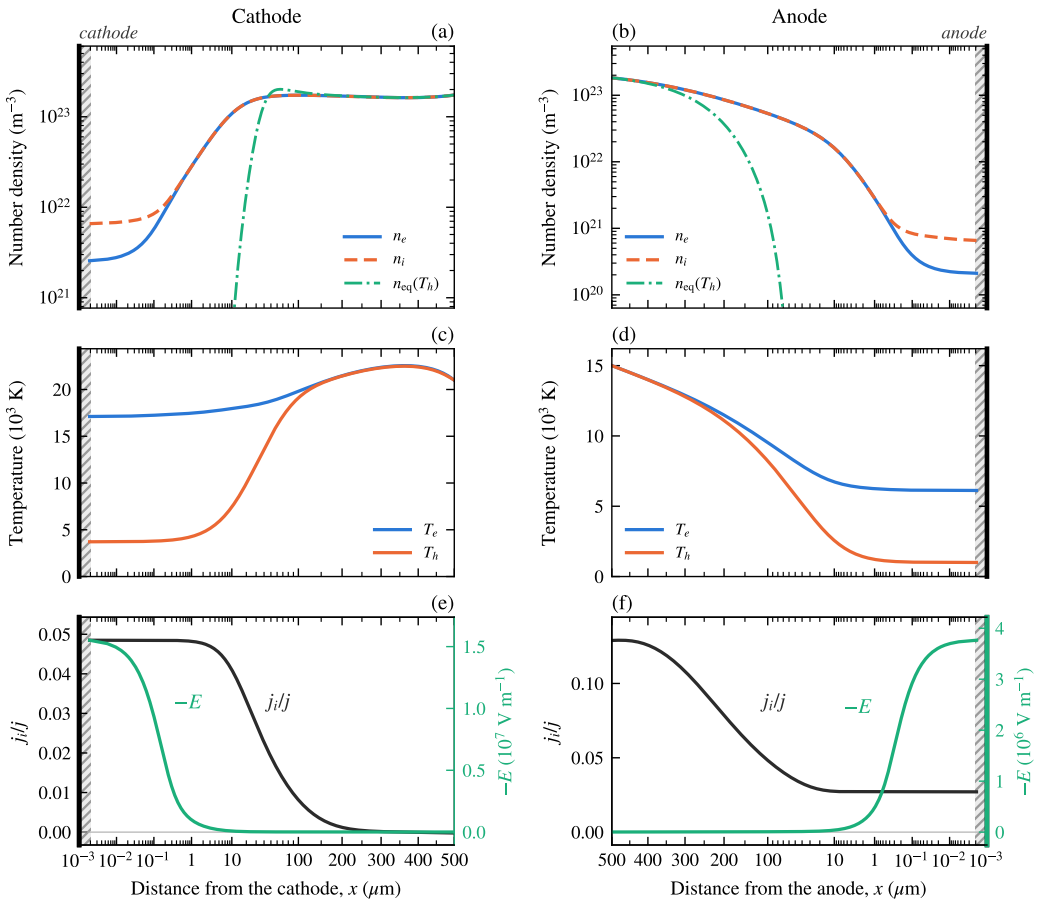


Figure 4. Steady profiles at the cathode (left) and the anode (right) against the distance x from the electrode. (a), (b) Number densities and the equilibrium density $n_{\mathrm{eq}}(T_h)$. (c), (d) Temperatures. (e), (f) j_i/j (left axis) and the field towards the electrode, $-E$ (right axis).

$x = 10^{-4}$ m, 4.2 kK at $x = 10^{-5}$ m and 5.1 kK at the surface, where $T_e = 6.1$ kK (figure 4(d)). The ions flow towards the anode across the whole layer, $j_i > 0$. They carry 2.7 % of the current at the surface, and their share grows towards the bulk plasma, to 13 % near $x = L$ (figure 4(f)).

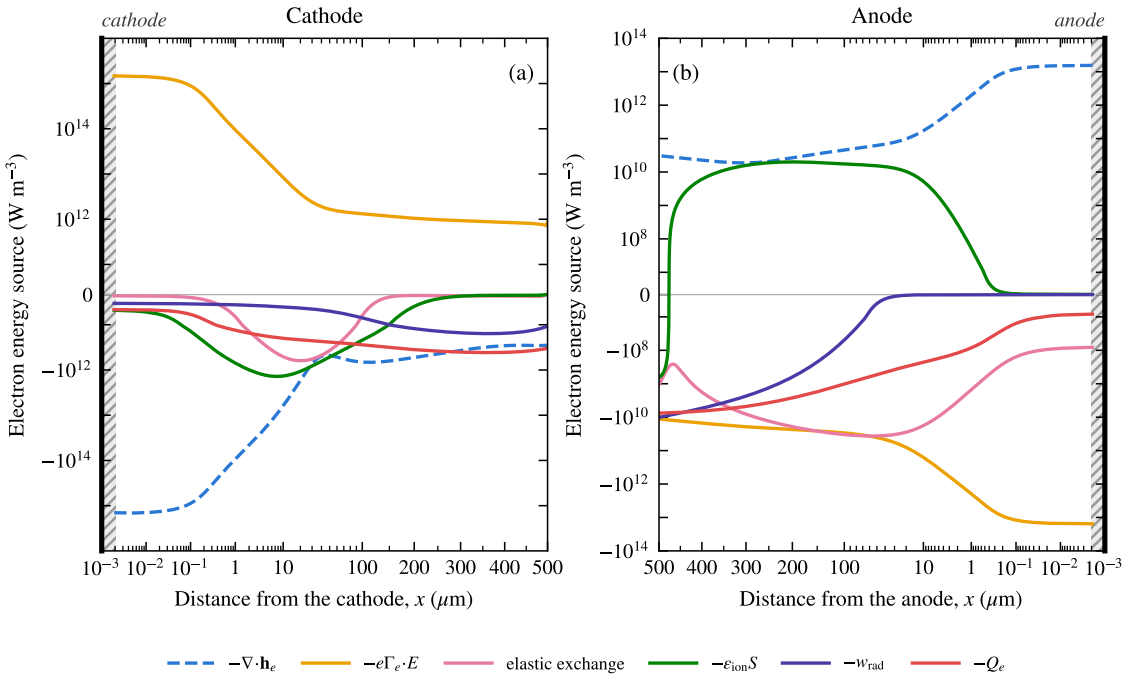


Figure 5. Terms of the steady electron energy balance (12) at (a) the cathode and (b) the anode, positive when they heat the electrons: $-\nabla \cdot \mathbf{h}_e$ (dashed) and the volume terms (solid).

4.2 Electron energy balance

Figure 5 shows every term of the steady electron energy equation (12). The divergence of the electron energy flux (13), $-\nabla \cdot \mathbf{h}_e$, is plotted as one term that combines the enthalpy flux, conduction and the thermal-diffusion heat flux. The volume terms are Joule heating $-e\boldsymbol{\Gamma}_e \cdot \mathbf{E}$, elastic exchange with the heavy particles, the ionization term $-\varepsilon_{\rm ion}S$, radiation $-w_{\rm rad}$ and the radial loss $-Q_e$ (14). Positive terms heat the electrons, and at steady state the terms sum to zero. The balance is very different at the two electrodes.

Cathode. Joule heating is the only source of electron energy, and it is the largest term everywhere (figure 5(a)). It peaks at the surface, 1.5×10^{15} W m^{-3}, where the sheath field accelerates the emitted electrons, and falls by more than two orders of magnitude within 10 μm. Over this distance it is balanced almost entirely by $-\nabla \cdot \mathbf{h}_e$, which takes up 97 % of the Joule heat released there. In the sheath alone, the two cancel to within 0.2 %, and the volume losses there are two to four orders of magnitude smaller than the Joule heating. The Joule heat released near the surface is therefore not lost locally but is carried by the electron energy flux to the cathode and towards the bulk. Beyond 10 μm, $-\nabla \cdot \mathbf{h}_e$ is small, and the Joule heating, now 10^{12}–10^{13} W m^{-3}, is balanced by the volume terms. Ionization is the largest of them out to about 90 μm, with a peak of 1.4×10^{12} W m^{-3} at 7 μm. The elastic exchange peaks at 6.2×10^{11} W m^{-3} at 20 μm, where it heats the heavy particles (figure 4(c)). Both terms become small beyond 200 μm, where T_h has approached T_e and the plasma is close to ionization equilibrium. In the column part of the domain, $x > 100$ μm, the radial loss, the electron energy flux towards $x = L$ and, to a lesser extent, radiation take up the Joule heating. Of the Joule heat released over the whole domain, 58 % leaves as electron energy flux into the column at $x = L$, and 12 % into the cathode with the absorbed plasma electrons. The radial loss takes 15 %, most of it at $x > 100$ μm, and ionization 7 %. Radiation takes 5 % and elastic exchange 3 %.

Anode. The Joule term is a loss throughout the domain (figure 5(b)), because the electrons flow towards the anode against the retarding field. Its magnitude is largest at the surface, 1.5×10^{13} W m^{-3}. Apart from recombination, the only source of electron energy is $-\nabla \cdot \mathbf{h}_e$, which is positive everywhere. Only 14 % of the electron energy flux that enters at $x = L$ reaches the surface. Within 10 μm of the anode, $-\nabla \cdot \mathbf{h}_e$ and the Joule loss balance each other to within 1 %, and this

region accounts for 45 % of the Joule loss over the domain. At the surface, the other volume terms are five orders of magnitude smaller. Further from the anode, the Joule loss decreases to about 10^{10} W m^{-3}, and the volume terms become comparable to it. The elastic loss to the heavy particles peaks at 3.7×10^{10} W m^{-3} at 57 μm, where $T_e - T_h$ is still 2.2 kK. The ionization term heats the electrons, with a peak of 2.0×10^{10} W m^{-3} at 199 μm. Here the plasma recombines, $S < 0$, because the charged particles are supplied by diffusion at densities well above the local ionization equilibrium (Section 3.2), and the recombination energy is returned to the electrons. Radiation and the radial loss grow towards the bulk and reach 1.0×10^{10} and 7.6×10^{9} W m^{-3} at $x = L$. Over the whole domain, $-\nabla \cdot \mathbf{h}_e$ supplies 80 % of the electron energy and recombination heating 20 %. Of this energy, the Joule loss takes 67 %, the elastic loss 23 %, the radial loss 6 % and radiation 4 %.

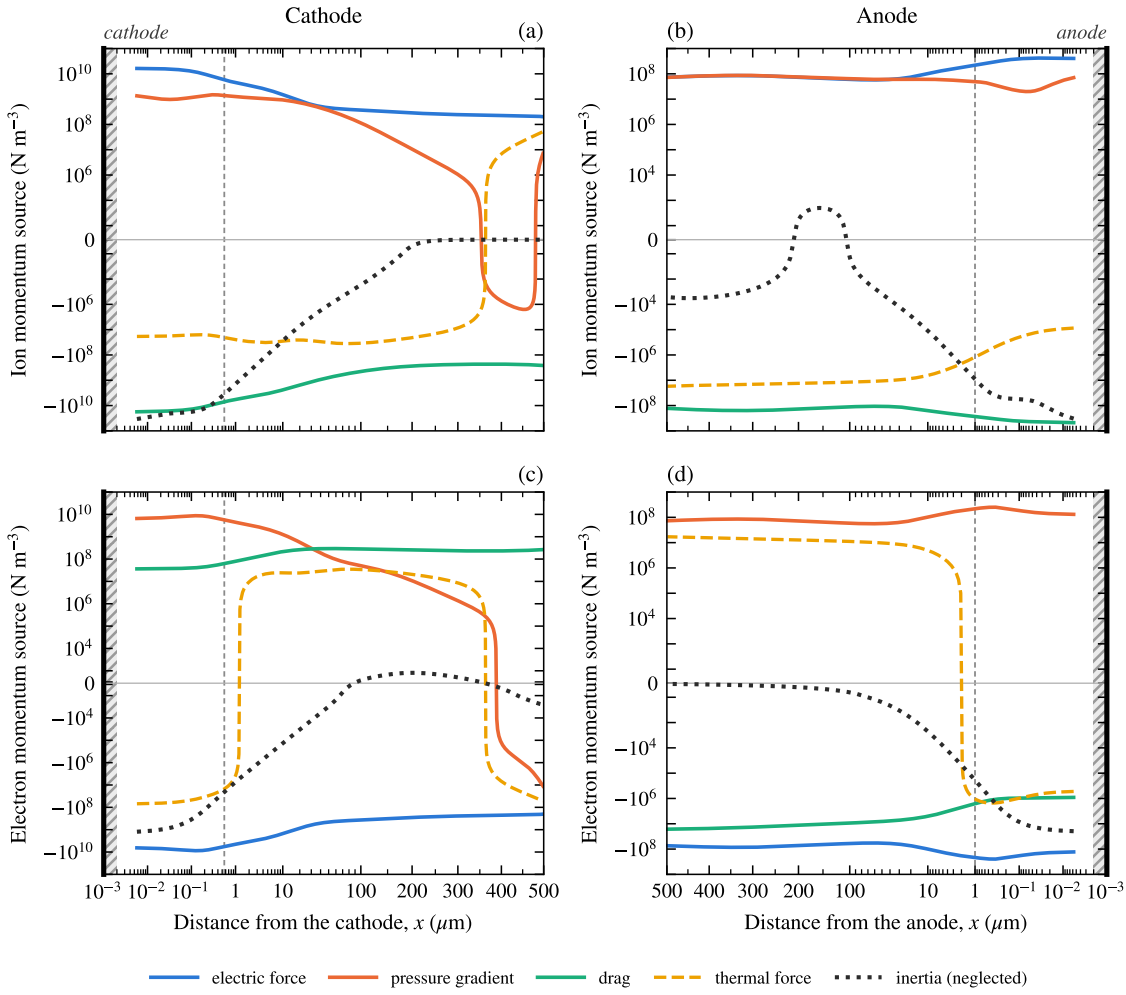


Figure 6. Terms of the steady momentum balances of the ions (4), (a), (b), and of the electrons (3), (c), (d), positive towards the electrode. Dashed vertical line: the sheath edge x_s.

4.3 Momentum balance and collisionality of the sheaths

The fluxes (9) and (10) follow from the force balances (3) and (4), in which the inertia of each species relative to the heavy gas is neglected. Figure 6 shows the terms of these balances across the domain in the two steady states, namely the electric force, the pressure gradient, the two collisional drags combined and the thermal force. In the steady states the heavy gas is at rest, so the share $Y_s\mathbf{F}$ of the net force is negligible. For comparison, the figure also shows the inertia $m_s n_s(\mathbf{u}_s \cdot \nabla)\mathbf{u}_s$ that the model neglects, evaluated a posteriori from the computed fluxes. The retained terms sum to zero to within 4×10^{-3} of the largest one.

Electrons. In both sheaths the field retards the electrons, and the electric force on them is balanced by their pressure gradient (figure 6(c), (d)). In the sheath the two cancel to within 0.9 % (cathode) and 1.3 % (anode) of the electric force. The drag and the thermal force are small there. The drag becomes important only in the quasi-neutral layer, where it carries the current. At 10 μm from the surface it is 14 % (cathode) and 5.0 % (anode) of the electric force, and at the anode the

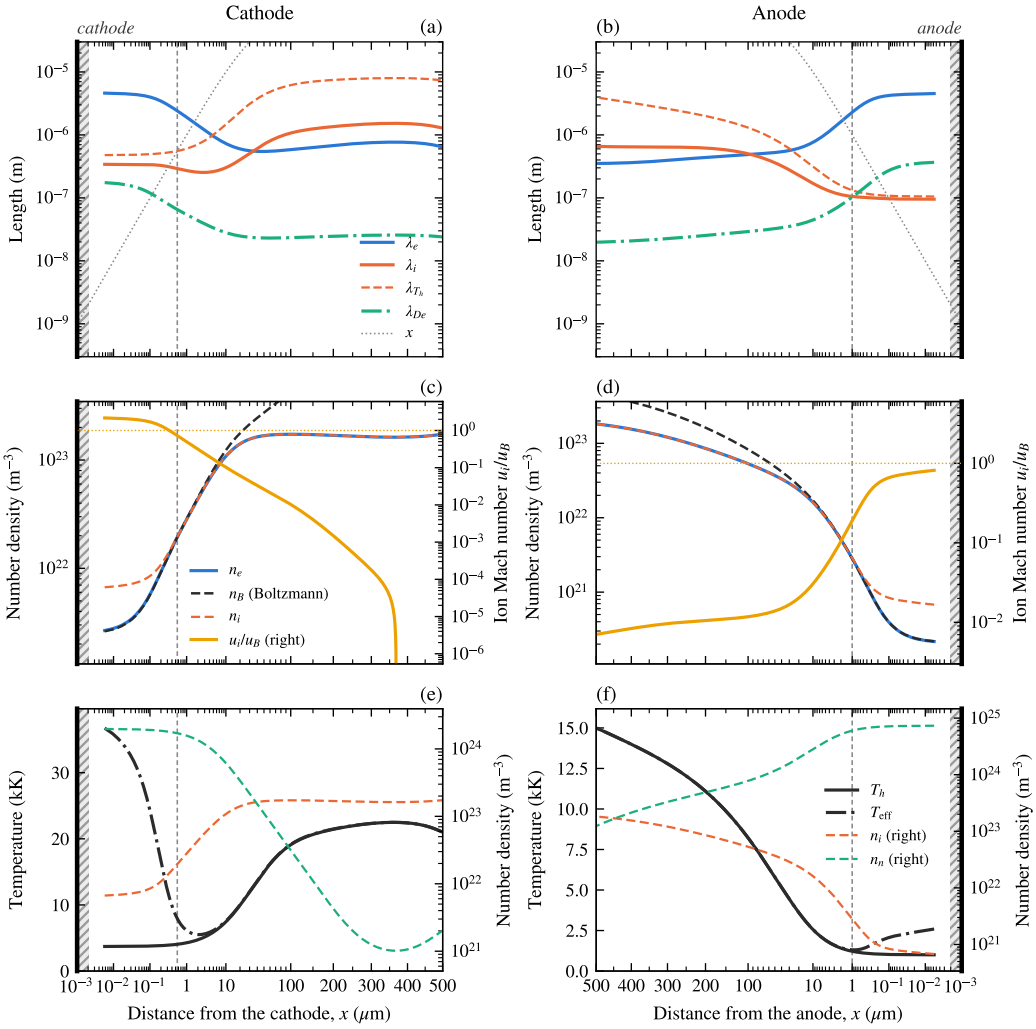


Figure 7. The sheaths at the cathode (left) and the anode (right). (a), (b) Mean free paths, the ion–atom energy-exchange length λ_{T_h} and the electron Debye length, with x dotted. (c), (d) n_e, the Boltzmann density n_B and n_i (left axis), and the ion Mach number u_i/u_B (right axis). (e), (f) T_h and the effective ion temperature $T_{\rm eff}$ (left axis), and n_i and n_n (right axis). Dashed vertical line: the sheath edge $x_{\rm s}$.

thermal force is as large there. The inertia of the electrons exceeds their drag within 0.33 μm of the cathode (by up to a factor of 34) and within 0.40 μm of the anode (by up to a factor of 22), but it remains small compared with the electric force and the pressure gradient, which dominate the balance. Figure 7(a), (b) shows the electron and ion mean free paths λ_e and λ_i of table 1, evaluated with the local densities and temperatures, together with the electron Debye length. The electron mean free path is of order microns, 2.4 μm at the cathode sheath edge and 2.4 μm at the anode one. The sheaths, 0.56 and 1.0 μm thick, therefore contain only 0.18 and 0.30 electron mean free paths, and the electrons cross both practically without collisions. Their balance in the sheaths is, however, not one of collisional transport. The drag is a few per cent of the electric force, and the field and the pressure gradient balance each other as in the Boltzmann equilibrium of a Maxwellian population retarded by a potential barrier, which does not require collisions. Simplified sheath models describe the electrons in the sheath by this Boltzmann relation, and figure 7(c), (d) tests it against the computed sheaths with $n_B = n_e(x_s)\exp[(e/k_B)\int_{\phi_s}^{\phi} \mathrm{d}\phi'/T_e]$, started at the sheath edge. The electron density follows the Boltzmann relation closely. The relation reproduces n_e to within 1.8 % across the cathode sheath, where n_e falls by a factor of 7.3, and to within 0.6 % across the anode sheath, where it falls by a factor of 13. Outside the anode sheath, it still holds to within 6 % up to 10 μm from the surface, but n_e is 16 % below n_B at 30 μm and 36 % below at 100 μm. At the cathode, it fails progressively in the ionization layer, where the electron drag carries the current, and n_e is 12 % below n_B at 10 μm and 59 % below at 100 μm. The electron description of Section 2.2 therefore remains adequate in the sheaths.

Ions. In both sheaths, the electric force on the ions is balanced by the drag (figure 6(a), (b)). At the cathode, the pressure gradient adds at most 24 % to the electric force in the sheath, and 8 % at the surface. At the anode, the field is weaker, and the pressure gradient, which also drives the ions towards the surface, is 17 % of the electric force at the surface and at most 22 % in the sheath. The thermal force on the ions is negligible, below 1 % of the drag in either sheath. The neglected inertia is not small inside the cathode sheath. It exceeds the drag within 0.27 μm of the surface, by a factor of 1.9 at the surface, stays above 10 % of the drag out to 1.9 μm, and falls below 1 % beyond 11 μm. At the anode, the inertia is at most 69 % of the drag (at the surface), and below 10 % beyond 0.45 μm. The ions are more collisional than the electrons. At the sheath edge (Figure 7(a), (b)), $\lambda_i = 0.29$ μm at the cathode and 0.11 μm at the anode, so that the ions cross 1.8 and 9.4 mean free paths in the sheaths. The anode sheath is thus collision-dominated for the ions, consistent with their inertia, which is below 10 % of the drag except within 0.5 μm of the surface. The cathode sheath is transitional, since the ions make one to two collisions across it and their inertia exceeds the drag over its inner half. There the ions cannot follow the steep rise of the field, and the local relation between field and ion velocity in equation (10) overestimates the ion velocity near the surface. Simplified sheath models impose the Bohm criterion on the ions entering the sheath, $u_i \geq u_B$, with $u_B = \sqrt{k_B(T_e + T_h)/M}$. The computed ions do not satisfy it at the sheath edge (figure 7(c), (d), right axis). At the cathode they enter the sheath at $u_i = 1540\ \mathrm{m\,s^{-1}}$, against $u_B = 2110\ \mathrm{m\,s^{-1}}$, i.e. at $0.73\,u_B$. They reach u_B only 0.33 μm from the surface, inside the sheath, where the charge separation is already 4.5 %, and arrive at the surface at $2.2\,u_B$. At the anode, the ions enter the sheath at $0.20\,u_B$ and reach the surface at $0.84\,u_B$, never attaining u_B. The Bohm flux $n_i u_B$ at the sheath edge would therefore overestimate the ion flux entering the sheath by 37 % at the cathode and by a factor of 5 at the anode. The Bohm criterion is derived for a collisionless sheath. In these sheaths the ions collide with the atoms throughout the presheath and the sheath, and, except in the inner part of the cathode sheath, their velocity is set by the balance of the electric force and the drag rather than by the potential drop through which they have fallen.

A single heavy-particle temperature. The model describes the ions and the atoms by one temperature T_h. This requires, first, that the two species exchange energy over a distance $\lambda_{T_h} = 1/[(n_i + n_n)\overline{Q}_{in}]$ (figure 7(a), (b)) short compared with the length $L_{T_h} = T_h/|\nabla T_h|$ over which T_h changes. This holds in both layers, where λ_{T_h}/L_{T_h} is at most 0.09 at the cathode (at the surface) and 0.025 at the anode. Second, the ions must not gain energy from the field faster than they share it with the atoms. The two-temperature closure (Appendix A) measures this heating by the effective ion temperature T_{eff}, which rises above T_h where the drift of the ions through the atoms is fast (figure 7(e), (f)). At the anode, the condition is satisfied except next to the surface, where T_{eff} is 2.6 times T_h. In the cathode sheath, the two temperatures diverge, and T_{eff} exceeds T_h by a factor of 2.0 at the sheath edge and of 9.9 at the surface. These values are, however, not accurate, because the closure assumes many collisions, whereas the ions cross the cathode sheath with only one to two. The single heavy-particle temperature model therefore is accurate throughout most of the near-electrode region, with minor deviations in the anode sheath and more pronounced deviations in the cathode sheath.

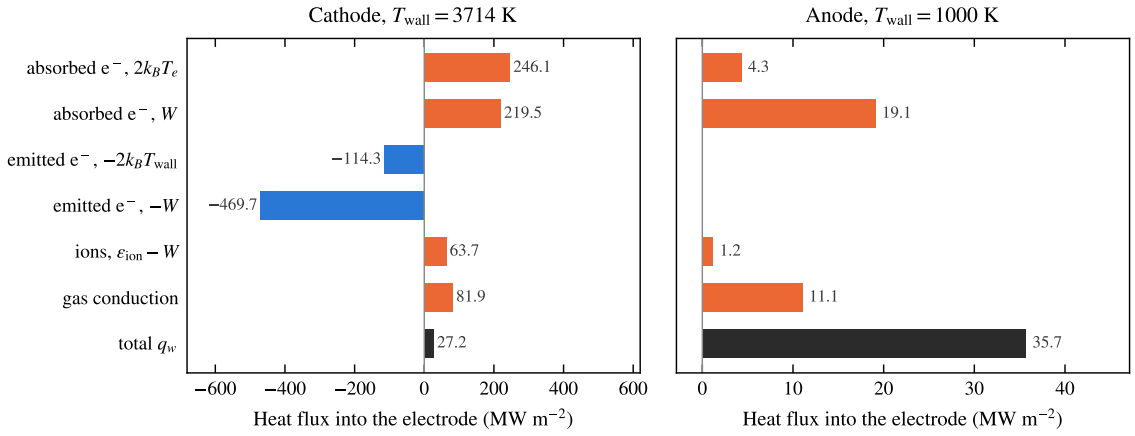


Figure 8. Terms of the heat flux into the cathode (left) and the anode (right), equation (24): heating (orange), emission cooling (blue) and the net heat flux q_w (black).

4.4 Heat flux to the electrode

The terms of the electrode heat flux (24) are shown in figure 8. At the cathode, the heat flux is the small difference of large terms. The absorbed plasma electrons (466 MW m^{-2}), gas conduction (82 MW m^{-2}) and ion recombination (64 MW m^{-2}) heat the surface by 611 MW m^{-2} in total, and emission cools it by 584 MW m^{-2}, so that only $q_w = 27.2$ MW m^{-2} is conducted into the cathode body. The surface temperature, 3714 K, is therefore fixed mainly by the condition that emission cooling nearly balances the heating, and only weakly by the conduction into the cathode body. Ion recombination, at $\varepsilon_{ion} - W = 13.1$ eV per ion, provides 10 % of the heating, although the ions carry only 4.8 % of the current. Equation (24) includes only the recombination energy of the ions. It assumes that their kinetic energy is returned to the neutrals at the surface. In industrial processes, however, the ion impact energy can play a significant role. We estimate the ion impact energy from the directed kinetic energy $\frac{1}{2}Mv_d^2$, using the ion drift velocity at the surface, $v_d = \Gamma_i/n_i = 4.6$ km s^{-1}. This corresponds to a drift velocity 2.2 times the local Bohm velocity u_B and an ion impact energy of 4.4 eV.

At the anode, $q_w = 35.7$ MW m^{-2}, of which the absorbed electrons deliver 66 %, 19.1 MW m^{-2} as the work function released on entering the metal and 4.3 MW m^{-2} as kinetic energy. Gas conduction contributes 11.1 MW m^{-2}, 31 %, and ion recombination 3.4 %. The anode heat flux is thus set by the electron current through W, by the electron temperature at the surface and, to a large part, by conduction in the heavy gas. Since the surface temperature is prescribed, this is the heat that the cooling of the anode removes. The ions carry 2.7 % of the current and fall through $\phi_s = 1.4$ V in a collision-dominated sheath, the ions reach the surface at $v_d = 1.0$ km s^{-1}, or $0.84\,u_B$ with an impact energy of 0.22 eV.

4.5 Dependence of the anode layer on the bulk temperature and the current density

On a thermionic cathode, the arc usually attaches in a spot with a high temperature and current density [10, 37]. The anode attachment, which is generally more important in plasma processing, can be either constricted or diffuse, depending on the arc current, the gas flow and the anode material [30]. The arc temperature and the current density in front of the anode vary depending on the attachment. To study the effect of these two parameters, the anode case of table 2 was repeated with the bulk temperature varied at fixed current density, $T_{bulk} = 13\,000$–$17\,000$ K at $J_{app} = 4 \times 10^6$ A m^{-2}, and with the current density varied at fixed bulk temperature, $J_{app} = 3 \times 10^6$–10^7 A m^{-2} at $T_{bulk} = 15\,000$ K (figure 9). The column radius follows the current density, $R = \sqrt{I/(\pi J_{app})}$ with $I = 200$ A. It is 3.99 mm over the temperature series and decreases from 4.61 to 2.52 mm over the current series. Figure 9(a)–(f) shows which profiles are set by the bulk temperature and which by the current.

Heavy-particle temperature. The T_h profile is set by T_{bulk} over the whole domain and is nearly independent of the current (figure 9(a), (b)). Over the current sweep, T_h varies by less than 3.0 %, whereas it increases with a increase in the bulk temperature.

Electron temperature. T_e is set by T_{bulk}, and is almost independent of current except within a few tens of micrometers from the anode. The current changes T_e by more than 4 % only within about 60 μm of the anode. At the surface, T_e falls from 6.6 to 5.2 kK as J_{app} increases from

3×10^6 to 10^7 A m^{-2}, whereas it rises from 5.0 to 7.1 kK between 13 000 and 17 000 K.

Charged-particle densities. The densities are set by T_{bulk} across the layer and by the current next to the anode (figure 9(c), (d)). At 10 μm from the anode, n_e rises by a factor of 3.5 between 13 000 and 17 000 K, from 7.5×10^{21} to 2.6×10^{22} m^{-3}, but changes by less than 6 % over the current sweep. Within a few micrometers of the anode, the order is reversed. The electrons that carry the current are absorbed at the thermal flux $\frac{1}{4} n_e \bar{c}_e$ (18), so that a larger current requires a larger electron density at the surface. There n_e rises by a factor of 3.6 with the current, from 1.5×10^{20} to 5.6×10^{20} m^{-3}, but decreases by 12 % with the bulk temperature. The ion density at the surface rises with both, by a factor of 2.9 with T_{bulk}, from 3.6×10^{20} to 1.0×10^{21} m^{-3}, and by 24 % with the current.

Potential and sheath. The potential depends on both parameters (figure 9(e), (f)). The voltage across the layer, $\phi(L)$, increases with the bulk temperature, from 4.0 to 5.9 V, and decreases with the current, from 5.6 to 3.9 V. The sheath drop ϕ_{s} follows the same trends, rising from 0.81 to 1.90 V with T_{bulk} and falling from 1.70 to 0.73 V with the current (figure 9(l), (n)). The sheath becomes thinner with both parameters (figure 9(k), (m)), from 1.12 to 0.91 μm with T_{bulk} and from 1.07 to 0.77 μm with the current, as the ion density at the surface rises.

Heat flux. The heat flux to the anode increases linearly with the current density, from 30.8 to 68.0 MW m^{-2} between 3×10^6 and 10^7 A m^{-2} (figure 9(i)). It has two parts of different origin. The electron terms are proportional to the current. The work function released by the absorbed electrons contributes 4.70–4.83 W per ampere, slightly more than W/e because the ions flow into the anode and the electron current exceeds the arc current by j_i, and their kinetic energy adds 0.91–1.19 W per ampere. Gas conduction, in contrast, does not depend on the current. It stays between 10.7 and 11.4 MW m^{-2} because the heavy-particle temperature profile is set by the bulk temperature. At fixed current density, q_w rises from 29.5 to 42.6 MW m^{-2} between 13 000 and 17 000 K (figure 9(g)). Most of this increase comes from gas conduction, which rises from 6.8 to 15.5 MW m^{-2}, and from ion recombination (0.4 to 2.4 MW m^{-2}). The electron terms rise by 11 %, with T_e at the surface rising from 5.0 to 7.1 kK.

Ion current. The ions flow into the anode in every case. Their current density at the surface hardly depends on the arc current. While the current density increases by a factor of 3.3, j_i stays between 0.95 and 1.19×10^5 A m^{-2} (figure 9(j)), so that the ion share of the current falls from 4.0 to 1.0 %. Like the heavy-particle temperature and the density of the layer, it is set instead by the bulk temperature. Between 13 000 and 17 000 K, j_i increases by a factor of 5.8, from 0.37×10^5 to 2.2×10^5 A m^{-2} (figure 9(h)), and the ion share of the current from 0.9 to 5.4 %.

In summary, the bulk temperature sets the heavy-particle temperature, the charged-particle density of the layer, the ion current and the gas conduction to the anode, while the current sets the electron density and, to a lesser extent, the electron temperature next to the anode, and the electron terms of the heat flux. The voltage across the layer, the sheath and the total heat flux depend on both.

4.6 Quasi-steadiness of the near-electrode layers

It is useful to know under which conditions the surface quantities can be considered quasi-steady, because this decides whether a transient model of the near-electrode region is needed or whether a steady-state model [12, 43, 14] can be coupled to an LTE model of the arc. Here we estimate how slowly the current and the bulk temperature must change for the ion current density j_i and the heat flux q_w at the surface to remain quasi-steady, that is, to follow the steady state at the instantaneous conditions to a given accuracy.

In the one-dimensional model, the pressure disturbances launched by a change of the current or of the bulk temperature can travel only along the axis, and they are reflected at the electrode, which delays the approach to the new steady state. In a real arc, however, these disturbances are relieved radially. We therefore remove the acoustic response in every case by multiplying the inertia of the heavy gas in equation (17) by $\epsilon = 10^{-4}$, which increases the speed of sound by a factor of $1/\sqrt{\epsilon} = 100$. This changes the steady-state quantities by less than 0.3 %.

The response of a surface quantity f, either j_i or q_w, to a step ΔX of $X = J_{\mathrm{app}}$ or T_{bulk} is measured in units of the step,

$$g(t) = \frac{f(t)/f_0 - 1}{\Delta X / X}, \tag{28}$$

where $\Delta X/X = 0.02$ is the relative size of the step and f_0 is the steady value before the step. The final value of g, denoted by g_∞, is the sensitivity of the steady state to X. The duration of the response, t_{s}, is the time after which $|g - g_\infty|$ stays below 0.01. Both are listed in table 3 for the two cases of table 2. We assume that the response is linear, so that g depends neither on the size nor on

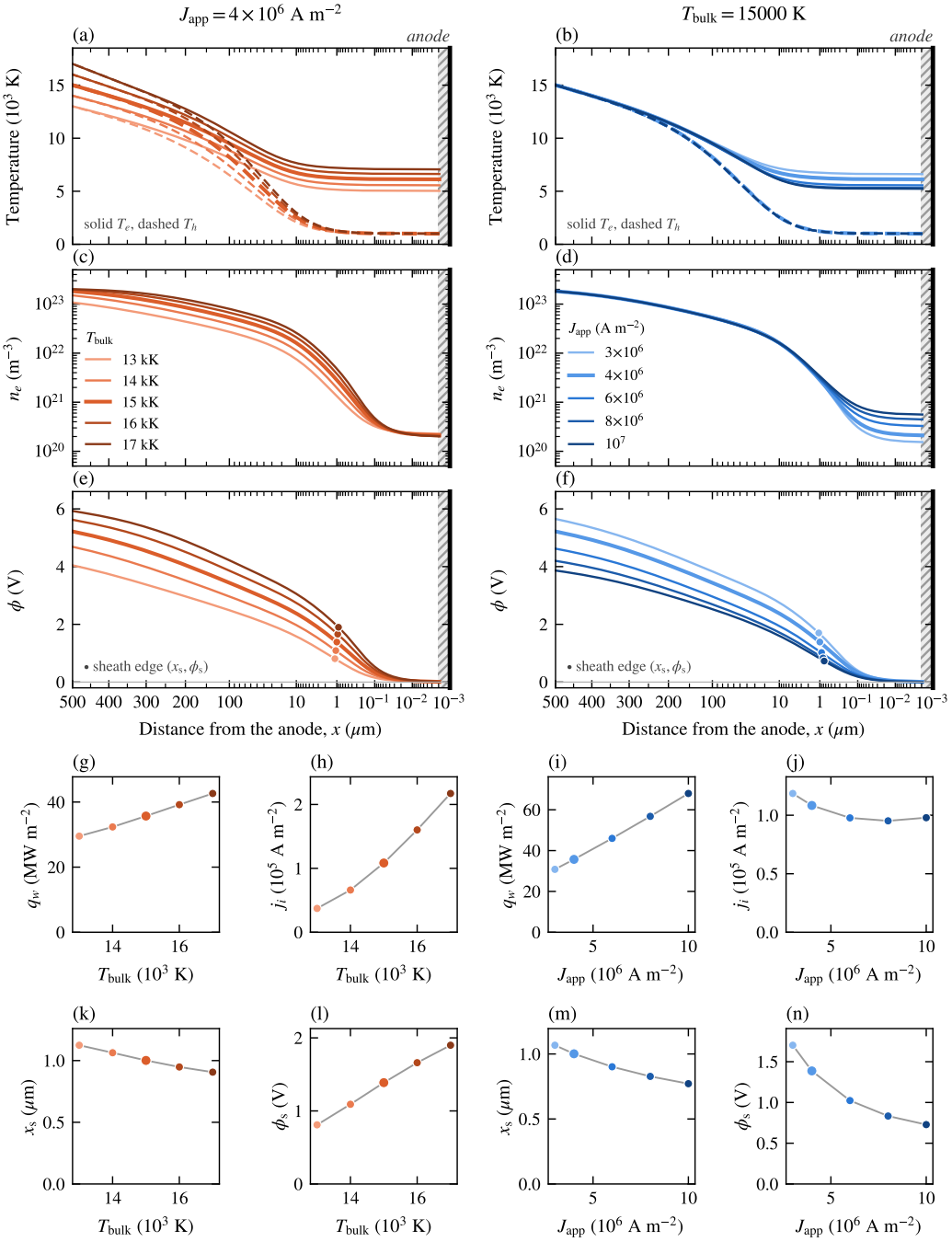


Figure 9. The anode layer for $T_{\rm bulk} = 13\,000$–$17\,000$ K at $J_{\rm app} = 4 \times 10^6$ A m^{-2} (left) and for $J_{\rm app} = 3 \times 10^6$–$10^7$ A m^{-2} at $T_{\rm bulk} = 15\,000$ K (right)(a), (b) T_e (solid) and T_h (dashed). (c), (d) n_e. (e), (f) Potential relative to the anode, with the sheath edge $(x_{\rm s}, \phi_{\rm s})$ marked. (g)–(n) Heat flux q_w, ion current density j_i, sheath width $x_{\rm s}$ and sheath drop $\phi_{\rm s}$.

the sign of the step. For the cases of table 2, steps of +2 and −2 % give values of the coefficient D_X defined below within 5 % of their mean.

Suppose now that X changes slowly, starting at $t = 0$ from a steady state in which $f = f_0$. Since the response is linear, it is the superposition of the responses to the small steps that make up the change,

$$\frac{f(t) - f_0}{f_0} = \int_0^t g(t - t') \frac{\mathrm{d} \ln X}{\mathrm{d}t'} \, \mathrm{d}t'. \tag{29}$$

A quasi-steady model instead gives the steady value at the instantaneous conditions,

$$\frac{f_{\mathrm{qs}}(t) - f_0}{f_0} = \int_0^t g_\infty \frac{\mathrm{d} \ln X}{\mathrm{d}t'} \, \mathrm{d}t'. \tag{30}$$

We subtract equation (30) from (29), substitute $s = t - t'$, and assume that $\mathrm{d} \ln X/\mathrm{d}t$ is nearly constant over the duration of the response. Once t exceeds t_s, the error of the quasi-steady approximation is then

$$\frac{f - f_{\mathrm{qs}}}{f_0} = -D_X \frac{\mathrm{d} \ln X}{\mathrm{d}t}, \qquad D_X = \int_0^\infty (g_\infty - g) \, \mathrm{d}s. \tag{31}$$

The coefficient D_X is the area between the step response and its final value. It measures how far f lags behind its quasi-steady value, and it includes the fast transients as well as the slow part of the response. We write D_J for $X = J_{\mathrm{app}}$ and D_T for $X = T_{\mathrm{bulk}}$. When both change at once, the two errors add, since the response is linear. The error stays within 1 % when X changes over a time $t_X = X/|\mathrm{d}X/\mathrm{d}t| \geq t_{1\%} = 100\,|D_X|$. Equation (31) also requires X to change little over the duration of the response, which is a few microseconds at the cathode and a few tens of microseconds at the anode (table 3).

Table 3. Response of the ion current density j_i and the heat flux q_w at the surface to steps of +2 % in J_{app} at fixed T_{bulk} and in T_{bulk} at fixed current, for the cases of table 2.

		Cathode		Anode	
		j_i	q_w	j_i	q_w
Step of J_{app}	g_∞	2.34	7.03	−0.24	0.57
	t_s (μs)	1.7	2.4	21	6.5
	D_J (μs)	0.45	3.4	−0.92	−0.19
	$t_{1\%}$ (μs)	45	340	92	19
Step of T_{bulk}	g_∞	0.013	0.122	6.8	1.47
	t_s (μs)	0.21	0.95	61	46
	D_T (μs)	0.006	0.045	79	16
	$t_{1\%}$ (μs)	0.6	4.5	7900	1600

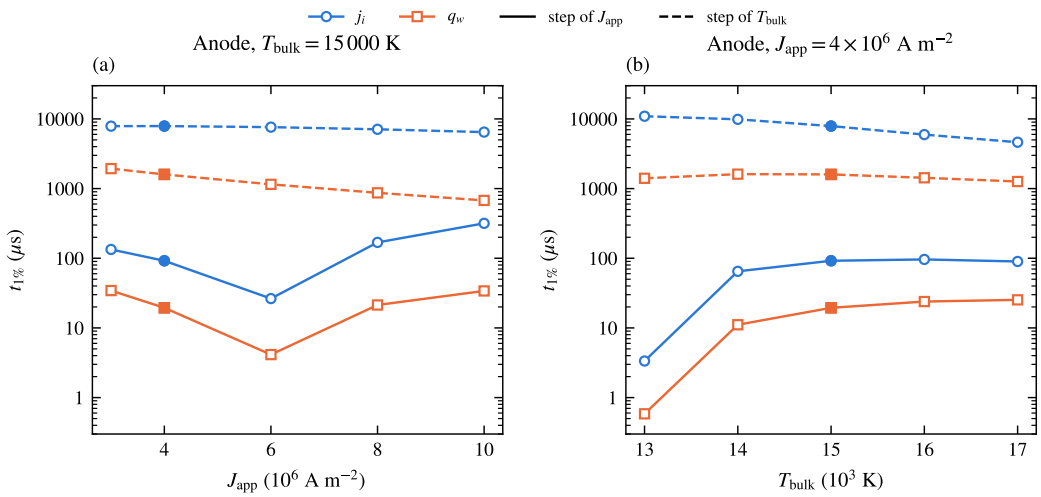


Figure 10. The time $t_{1\%} = 100\,|D|$ over which J_{app} (solid) or T_{bulk} (dashed) must change for j_i (blue) and q_w (orange) to stay within 1 % of their quasi-steady values at the anode, (a) against J_{app} and (b) against T_{bulk}. Filled symbols mark the case of table 2.

In every case below, the bulk densities follow their equilibrium values $n_{\mathrm{eq}}(T_{\mathrm{bulk}})$ after the step, and the case is integrated to the new steady state with R and T_{wall} held at their steady values.

Cathode. The cathode case of table 2 was perturbed by a step of +2 % in J_{app} at fixed T_{bulk} and, separately, by a step of +2 % in T_{bulk} at fixed current. After the current step, the ion current density rises by 4.7 %, with $D_J = 0.45$ μs, so that $t_{1\%} = 45$ μs. The heat flux first falls, by up to 31 % within 10 ps, and then rises to 14 % above its initial value. Because this rise is slow and is seven times the relative change of the current, q_w has a larger coefficient, $D_J = 3.4$ μs, and $t_{1\%} = 0.34$ ms (table 3). At fixed current, the cathode layer is almost insensitive to the bulk temperature. A step of 2 % in T_{bulk} changes j_i by only 0.03 % and q_w by 0.2 %. Accordingly, D_T is small, 6 ns for j_i and 45 ns for q_w, and $t_{1\%}$ is 0.6 and 4.5 μs, respectively.

Anode. The same steps were applied to the anode case of table 2, whose response parameters are listed in table 3. After the current step, the heat flux, which is dominated by the electron terms, follows the current almost instantly and rises by 1.1 %, and only a small part of its change is left to the slow part of the response. The ion current density falls by only 0.5 %, mostly slowly. Both coefficients D_J are therefore small, -0.19 μs for q_w and -0.92 μs for j_i. In comparison to the cathode, the anode layer depends strongly on the bulk temperature (Section 4.5). A step of 2 % in T_{bulk} raises j_i by 14 % and q_w by 3 %, and the response is much slower than at the cathode. The coefficients D_T are therefore nearly two orders of magnitude larger than D_J, so that at the anode it is the bulk temperature, rather than the current, that limits the quasi-steady approximation. The same steps were applied to every case of the anode sweeps of Section 4.5, and figure 10 shows the resulting $t_{1\%}$.

Figure 10 summarizes the sweeps. In every case, $t_{1\%}$ is more than an order of magnitude longer for the temperature step (dashed lines) than for the current step (solid lines), so that the bulk temperature sets the limit for the quasi-steady approximation over the whole range. For the current step, $t_{1\%}$ has a minimum near 6×10^6 A m^{-2}, where D_J changes sign, and becomes much shorter at low bulk temperatures, where the slow part of the response almost vanishes. For the temperature step, $t_{1\%}$ of the ion current density decreases as the bulk temperature rises and changes little with the current, while that of the heat flux decreases with the current and changes little with the bulk temperature.

Both layers can therefore be treated as quasi-steady, to within 1 % with respect to electrode ion current j_i and heat flux q_w, when the current changes over more than about 0.35 ms. The bulk temperature is irrelevant at the cathode, but at the anode it must change over more than about 2 ms for the heat flux and 11 ms for the ion current to remain quasi-steady. In a typical AC arc at 50 or 60 Hz, the half-period is about 10 ms. If the bulk temperature stays nearly constant over the cycle, both layers can be considered quasi-steady except within about 0.35 ms of zero current. If the bulk temperature follows the current, the cathode layer is unaffected where as anode layer can no longer be treated as quasi-steady.

5 Conclusion

A unified one-dimensional model of the near-electrode layers of an atmospheric-pressure argon arc has been presented. A single set of equations is solved from the electrode surface, through the space-charge sheath and the ionization layer, to a bulk-plasma boundary whose state is supplied by an LTE model of the arc or by measurements. The model includes separate electron and heavy-particle energy equations, thermal diffusion and the collisional drag between species, a heavy-gas momentum equation, the radial heat loss of the column and, at the cathode, a surface heat balance with thermionic emission.

The main predictions of the model for the cathode and anode layers of a high-intensity atmospheric-pressure arc are as follows.

- The two layers differ in structure. At the cathode, a thin ion-rich sheath with a voltage of 3.1 V is followed by an ionization layer of about 10^{-4} m, in which the hot electrons heat the gas and ionize the plasma almost fully. At the anode, the sheath is also ion-rich but weak, 1.4 V, and most of the 5.2 V retarding drop lies in the quasi-neutral layer. There the charged particles, supplied by diffusion, exceed their equilibrium density, and the electron and heavy-particle temperatures separate strongly.

- The electrons cross both sheaths almost without collisions, yet the drift-diffusion description of them remains adequate, because their momentum balance is dominated by the pressure gradient and the electric force. The ions are collisional and do not reach the Bohm velocity at the sheath edge, so that a Bohm flux would overestimate the ion flux by 37 % at the cathode and by a factor of 5 at the anode.

- The cathode heat flux, 27 MW m^{-2}, is the small difference between the heating of the surface and emission cooling, and it is therefore set mainly by the electrode rather than by the plasma. The anode heat flux, 36 MW m^{-2}, is two-thirds electronic and one-third gas conduction.
- At the anode, the bulk temperature sets the heavy-particle temperature, the charged-particle density, the ion current and the gas conduction. The current sets the electron density at the surface and the electron part of the heat flux, so that the heat flux increases linearly with the current. The sheath drop and the sheath width depend on both.
- Both layers are quasi-steady to within 1 % when the current changes over more than about 0.35 ms. The cathode layer is almost insensitive to the bulk temperature, whereas the anode layer is quasi-steady only when the bulk temperature changes over more than about 2 ms for the heat flux and 11 ms for the ion current.

Future work will extend the model to two dimensions and couple it to an LTE model of the arc. An axisymmetric model would resolve the radial structure of the electrode attachment, so that constricted and diffuse anode attachments can be compared. Coupled to an LTE model of the bulk plasma, would enable us to determine the species fluxes, the ion current and the heat transfer to the electrodes self-consistently together with the arc dynamics.

A Transport coefficients

This appendix gives the collision cross sections, describes how the transport and energy-exchange coefficients that close the model of Section 2.2 are calculated from them, and compares the resulting conductivities with the transport properties of argon in LTE.

A.1 Collision cross sections

The electron–atom collisions are described by the energy-dependent momentum-transfer cross section $Q^{(1)}(\varepsilon)$ of argon from the LXCat database [44], where ε is the electron energy. The average cross sections, denoted by an overbar, are averages over the Maxwellian energy distribution of the colliding particles. Following Almeida et al. [12], the average electron–atom and electron–ion momentum-transfer cross sections over the electron energy distribution are

$$\overline{Q}_{en} = \frac{1}{2}\int_0^\infty x^2 e^{-x}\, Q^{(1)}(x\, k_B T_e)\, dx, \qquad \overline{Q}_{ei} = \frac{e^4 \ln\Lambda}{32\pi\varepsilon_0^2 (k_B T_e)^2}, \tag{32}$$

where $x = \varepsilon/(k_B T_e)$ is the reduced electron energy. The second expression is the same average of the Coulomb momentum-transfer cross section. They are related to the energy-exchange frequencies by $\bar{\nu}_{en}^E = \frac{4}{3} n_n \bar{C}_e \overline{Q}_{en}$ and $\bar{\nu}_{ei}^E = \frac{4}{3} n_i \bar{C}_e \overline{Q}_{ei}$, equations (41) and (43), with $\bar{C}_e = \sqrt{8 k_B T_e/(\pi m_e)}$ the mean electron speed. Here $\ln\Lambda$ is the Coulomb logarithm,

$$\ln\Lambda = \ln\frac{\lambda_{De}}{b_{90}}, \qquad \lambda_{De} = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_e e^2}}, \qquad b_{90} = \frac{e^2}{12\pi\varepsilon_0 k_B T_e}, \tag{33}$$

where λ_{De} is the electron Debye length and b_{90} is the 90°-deflection impact parameter.

The ion–atom collisions are governed by resonant charge exchange. Their average momentum-transfer cross section over the energy distribution at the heavy-particle temperature is taken from the fit [38]

$$\overline{Q}_{in} = 2\left[7\times10^{-10} - 0.6\times10^{-10}\ln\left(\frac{2k_B T_h}{e}\right)\right]^2 \text{ m}^2, \tag{34}$$

with $2k_B T_h/e$ in eV. The average electron–electron cross section $\overline{Q}_{ee}$ is $\overline{Q}_{ei}$, and the average ion–ion cross section $\overline{Q}_{ii}$ is $\overline{Q}_{ei}$ evaluated at T_h.

The scales of table 1 are evaluated with these cross sections in LTE at atmospheric pressure, $T_e = T_h = T$, with the composition given by the Saha equation, quasi-neutrality and Dalton's law. The reaction length λ_{rxn} uses the ambipolar diffusion coefficient

$$D_a = \left(1 + \frac{T_e}{T_h}\right) D_{in}, \qquad D_{in} = \frac{3\pi}{32}\frac{\bar{C}_{in}}{(n_i + n_n)\,\overline{Q}_{in}}, \qquad \bar{C}_{in} = \sqrt{\frac{8 k_B T_h}{\pi\, m_{in}}}, \tag{35}$$

where D_{in} is the ion–atom diffusion coefficient [12] and $m_{in} = M/2$ is the ion–atom reduced mass, and the reaction time τ_{rxn} uses the rate coefficients of Almeida et al. [12], from Benilov and Naidis [38].

A.2 Electrons

Following Mitchner and Kruger [45], the electron current density $\mathbf{J}_e = -e\boldsymbol{\Gamma}_e$ and the electron heat flux $\mathbf{q}_e$ are written as

$$\mathbf{J}_e = \sigma\left(\mathbf{E} + \nabla p_e/(en_e)\right) + \varphi\,\nabla T_e, \tag{36}$$

$$\mathbf{q}_e = -\lambda \nabla T_e + \psi\,\mathbf{J}_e. \tag{37}$$

Here σ is the electrical conductivity, φ is the thermal-diffusion coefficient, λ is the ordinary (zero-current) thermal conductivity, and ψ is the energy transported by the current per unit charge. Comparing equations (36) and (37) with the electron momentum balance (3) and the energy flux (13) gives

$$\bar{\nu}_e^M = \frac{n_e e^2}{m_e \sigma}, \qquad C_e = \frac{e}{k_B}\frac{\varphi}{\sigma}, \qquad \kappa_e = \lambda, \qquad -\frac{e\psi}{k_B T_e} = \frac{5}{2} + C_e. \tag{38}$$

Mitchner and Kruger [45] give σ, φ, and λ in the weakly and fully ionized limits. Substituting them into equation (38) gives the collision frequencies and thermal-diffusion coefficients in each limit.

Weakly ionized limit. Electron–neutral collisions dominate, and the electrons behave as a Lorentz gas whose distribution function has a Maxwellian isotropic part $f_M(T_e)$. Let C be the electron speed, $\varepsilon = \frac{1}{2}m_e C^2$ the electron energy, $x = \varepsilon/(k_B T_e)$ the reduced energy, and $\nu_{en} = n_n C Q^{(1)}(\varepsilon)$ the energy-dependent momentum-transfer frequency, where $Q^{(1)}$ is the electron–atom momentum-transfer cross section (Appendix A.1). A two-term solution of the electron Boltzmann equation then gives

$$\bar{\nu}_{en}^M = \frac{\langle C^2\rangle_0}{\langle C^2/\nu_{en}\rangle_0}, \qquad C_{en} = \langle x\rangle_\nu - \frac{5}{2}, \qquad \lambda_{en} = \frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{en}^M}\left[\langle x^2\rangle_\nu - \langle x\rangle_\nu^2\right], \tag{39}$$

where $\bar{\nu}_{en}^M$ is the momentum-transfer average of ν_{en}, $\langle\cdot\rangle_0$ is the Maxwellian average, and $\langle\cdot\rangle_\nu$ is the transport average with the weight $C^4 f_M/\nu_{en}$,

$$\langle g\rangle_0 = 4\pi\int_0^\infty g\,C^2 f_M\,dC, \qquad \langle g\rangle_\nu = \frac{\langle g\,C^2/\nu_{en}\rangle_0}{\langle C^2/\nu_{en}\rangle_0}. \tag{40}$$

The elastic energy exchange in equations (12) and (15) uses instead the energy-exchange average

$$\bar{\nu}_{en}^E = \frac{\langle C^2\nu_{en}\rangle_0}{\langle C^2\rangle_0}. \tag{41}$$

Fully ionized limit. For singly charged ions, the Spitzer–Härm solution gives [45]

$$\bar{\nu}_{ei}^M = \frac{\bar{\nu}_{ei}^E}{1.975}, \qquad C_{ei} = \frac{1.389}{1.975} = 0.703, \qquad \lambda_{ei} = 3.203\,\frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{ei}^E}, \tag{42}$$

where $\bar{\nu}_{ei}^E$ is the energy-exchange average (41) of the Coulomb collision frequency [45],

$$\bar{\nu}_{ei}^E = n_i\,\frac{4\sqrt{2\pi}}{3}\left(\frac{m_e}{k_B T_e}\right)^{3/2}\left(\frac{e^2}{4\pi\varepsilon_0 m_e}\right)^2 \ln\Lambda, \tag{43}$$

and $\ln\Lambda$ is the Coulomb logarithm, equation (33).

At intermediate ionization, the two limits are combined. The momentum-transfer frequencies are added, the thermal conductivities are combined in series (their thermal resistances add), and the thermal-diffusion coefficient is the mean of the two limits weighted by the energy-exchange frequencies,

$$\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M, \qquad \frac{1}{\kappa_e} = \frac{1}{\lambda_{en}} + \frac{1}{\lambda_{ei}}, \qquad C_e = \frac{C_{en}\bar{\nu}_{en}^E + C_{ei}\bar{\nu}_{ei}^E}{\bar{\nu}_{en}^E + \bar{\nu}_{ei}^E}. \tag{44}$$

A.3 Ions

The ion–neutral transport is governed by resonant charge exchange. The low-field mobility is obtained from the ion–atom cross section (34) with the Mason–Schamp equation [46],

$$\mu_i^0 = \frac{1.104\times 10^5}{\overline{Q}_{in}\sqrt{T_h}\,n_n}\ \mathrm{m^2\,V^{-1}\,s^{-1}} \qquad (\overline{Q}_{in}\ \text{in m}^2,\ T_h\ \text{in K},\ n_n\ \text{in m}^{-3}). \tag{45}$$

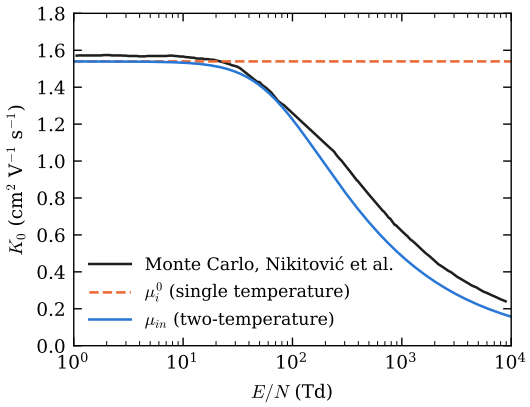


Figure 11. Reduced mobility of Ar^+ in Ar at 300 K: single-temperature μ_i^0 (45) (dashed), two-temperature μ_{in} (46), (47) with $F_{in} = |\mathbf{E}|$ (solid blue), and the Monte Carlo results of Nikitović and Raspopović [49].

In strong fields the ions are heated by their drift through the atoms. This is accounted for by the two-temperature theory of Wannier [47, 48], in which the drift of the ions relative to the atoms, $v_d = |\mathbf{u}_i - \mathbf{u}_n|$, raises their random energy to an effective temperature T_{eff}, which sets the ion–atom mobility,

$$T_{\text{eff}} = T_h + \frac{M v_d^2}{3k_B}, \qquad \mu_{in} = \mu_i^0 \sqrt{\frac{T_h}{T_{\text{eff}}}}. \tag{46}$$

In the ion momentum balance (4), the friction with the atoms, $M n_i \bar{\nu}_{in}^M (\mathbf{u}_i - \mathbf{u}_n)$, balances the remaining forces on the ions, a force eF_{in} per ion with

$$F_{in} = \frac{1}{e n_i} \left| e n_i \mathbf{E} - \nabla p_i - M n_i \bar{\nu}_{ie}^M (\mathbf{u}_i - \mathbf{u}_e) + C_i n_e k_B \nabla T_e - Y_i \mathbf{F} \right|.$$

Hence $v_d = \mu_{in} F_{in}$, and with equation (46) the drift speed follows in closed form,

$$v_d^2 = \frac{\sqrt{1 + 4\mathcal{A}\mathcal{B}} - 1}{2\mathcal{B}}, \qquad \mathcal{A} = \left(\mu_i^0 F_{in}\right)^2, \qquad \mathcal{B} = \frac{M}{3k_B T_h}. \tag{47}$$

For ions drifting in a uniform field through a gas at rest, $F_{in} = |\mathbf{E}|$, and equations (46) and (47) take the form in which the closure is usually written, with $v_d = \mu_{in}|\mathbf{E}|$ [47]. The ion–neutral momentum-transfer frequency is then obtained from the mobility as

$$\bar{\nu}_{in}^M = \frac{e}{M \mu_{in}}. \tag{48}$$

Figure 11 compares the reduced mobility $K_0 = \mu n_n / N_0$ (with $N_0 = 2.687 \times 10^{25}$ m^{-3}) of the single-temperature mobility μ_i^0, equation (45), and the two-temperature mobility μ_{in}, equations (46) and (47) with $F_{in} = |\mathbf{E}|$, with the Monte Carlo results of Nikitović and Raspopović [49] for Ar^+ in Ar at 300 K. The two-temperature mobility reproduces the decrease of K_0 with E/N and lies 15 % below the Monte Carlo value at 300 Td and 26 % below at 3000 Td.

The ion–electron frequency follows from momentum conservation, equation (6),

$$\bar{\nu}_{ie}^M = \frac{m_e n_e}{M n_i} \bar{\nu}_{ei}^M, \tag{49}$$

with $\bar{\nu}_{ei}^M$ given by equation (42). The ions take up the electron–ion part of the electron thermal force, so their thermal coefficient is the C_{ei} term of equation (44),

$$C_i = \frac{C_{ei} \bar{\nu}_{ei}^E}{\bar{\nu}_{en}^E + \bar{\nu}_{ei}^E}. \tag{50}$$

The ion contribution κ_i to the heavy-particle thermal conductivity is calculated as in Almeida et al. [12].

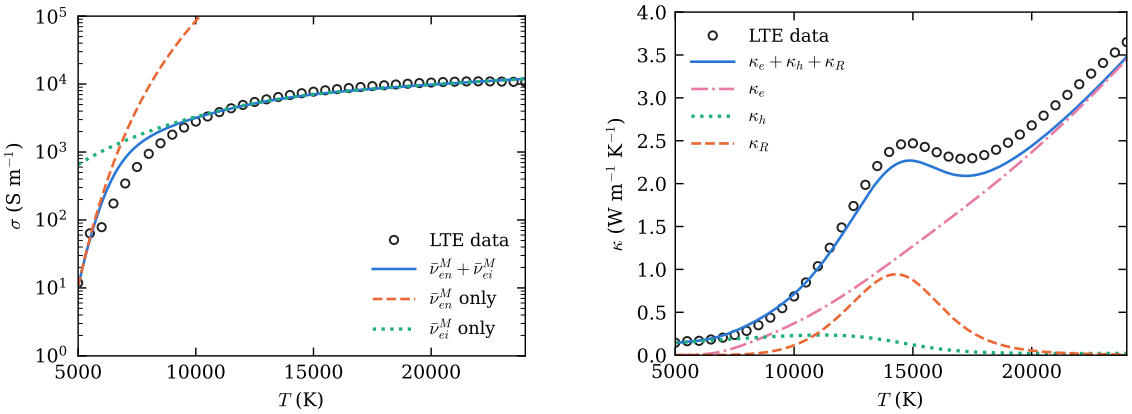


Figure 12. Transport coefficients of argon in LTE at atmospheric pressure from equation (44) (lines) and from Boulos et al. [1] (symbols). Left: electrical conductivity, with its electron–neutral and electron–ion limits. Right: thermal conductivity $\kappa_e + \kappa_h + \kappa_R$, with κ_h from [12] and κ_R from Mutation++ [50].

A.4 Heavy species

The atom frequencies follow from momentum conservation, equation (6),

$$\bar{\nu}^M_{ni} = \frac{n_i}{n_n}\,\bar{\nu}^M_{in}, \qquad \bar{\nu}^M_{ne} = \frac{m_e n_e}{M n_n}\,\bar{\nu}^M_{en}. \tag{51}$$

The atoms take up the electron–neutral part of the electron thermal force, so their thermal coefficient is the C_{en} term of equation (44),

$$C_n = \frac{C_{en}\bar{\nu}^E_{en}}{\bar{\nu}^E_{en} + \bar{\nu}^E_{ei}}, \tag{52}$$

and equations (50) and (52) satisfy $C_i + C_n = C_e$, as required by equation (6). The heavy-particle thermal conductivity $\kappa_h = \kappa_n + \kappa_i$, the sum of the atom and ion contributions, is calculated as in Almeida et al. [12].

A.5 Comparison with LTE transport properties

The electrical conductivity $\sigma = n_e e^2/(m_e \bar{\nu}^M_e)$ and the thermal conductivity obtained with these coefficients are compared with the transport properties of argon in local thermodynamic equilibrium (LTE) in figure 12. The coefficients are evaluated with $T_e = T_h = T$ and the Saha composition at atmospheric pressure, and compared with the data tabulated by Boulos et al. [1]. The reactive conductivity κ_R in our model resides in the ionization term and therefore for the comparison, the reactive conductivity computed with Mutation++ [50] is added to $\kappa_e + \kappa_h$.

The electrical conductivity agrees well with the LTE data over most of the temperature range, with only minor deviations at low temperatures, where electron–neutral collisions dominate. At intermediate temperatures, however, the additive rule for the collision frequencies overestimates the conductivity. In argon, the slow electrons near the Ramsauer minimum carry a substantial part of the current in the electron–neutral limit, but these same electrons are strongly scattered by the ions, and adding the averaged electron–neutral and electron–ion collision frequencies does not capture this correlation between the two collision processes. The thermal conductivity also agrees well with the reference data at high temperatures, and is moderately overpredicted at intermediate temperatures.

Acknowledgments

To be completed.

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